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Abstract:

Information Centric Networking (ICN) is a new network paradigm for future Internet approach aiming to intercept the architectural problems present in current Internet model and incompetence by reciprocating the main unit of network from hosts to content itself. In ICN model, the content names are crucial. Every content has been provided with a unique name over the network layer and this name further helps in routing within the network. ICN focuses on content retrieval regardless of where the content has been located. Therefore, in ICN instead of securing the infrastructure or the network entities it is much more important to secure the content. This new paradigm has also bought certain new security challenges as there is no existence of any point to point or client server authentication mechanism. Information Centric Networking (ICN) is a novel future Internet model approach to the network that attempts to preempt the architectural issues inherent in the current Internet model and incompetence that attempts to reciprocate the primary unit of network, the hosts, with the content itself. This thesis work is projected in the hope that it can assist the research community, by giving an abstract notion about the security issues in ICN model. To achieve the security targets in ICN, it is certainly important to have deep understanding of security attacks that could exists in this model, their classification as well as the mitigation strategies. This thesis work performs a survey and summarizes the existing attacks that have impact on ICN by classifying them into four main categories namely; Name based, Routing based, Caching based and multifarious attacks. It also provides the counter measures to mitigate these security attacks. Later on, in this work a relationship has been presented between ICN attributes and the security attacks. A brief introduction of existing Access Control mechanism has also been provided that could relate the security needs. This thesis work has been presented with the aim of helping research community by providing an overall idea about the security concerns in ICN model.

Chapter 1

Introduction

1.1 Introduction:

During the current period, networks are being utilized more for content retrieval than for straight communication between two people or entities (C. Annual and I. Report, 2018). This mode urges a new network architecture particularly for the retrieval of content. Various solutions have been proposed to meet to users demand and realize the Future Internet Architecture (FIA). The End-users are more concerned in what they are requesting for and consuming irrespective of from where the content or services are coming. As a result, there came an emergence of new network paradigm named as Information-Centric Networking. In contrast the current internet which is basically a host-centric model and it uses an IP address to convey the packets towards the destination host. ICN only uses name of the content to carry out the communication and yields the content from the network (Nou and Moungla, 2021). With the help of naming contents over the network layer, the feature of in-network caching mechanism is natively supported by ICN (Zhang and Luo, 2015). This feature leads to facilitating efficient usage of network resources and timed delivery of the content to its requesters. In this regard, Information- Centric Networking has unfolded as a favorable candidate for the Future Internet paradigm. Considering the fact that the Internet is widely being used to information dissemination rather than peer-to-peer communication, ICN focused to fulfill the current and future demands and compared to the existing host-centric internet architecture. ICN supports in-network caching mechanism (precisely storage) by naming the information over network layer. Thus, this feature promises the timed and efficient delivery of the information to the requesters. We are aware of the fact that in IP based networks, the IP addresses are associated with a network entity (a server or any node). Apart from information distribution, ICN has segregated various limitations that are present in the current networking architecture. For instance, end user mobility, security enforcement as it secures each named data object with the content credibility rather than securing the point-to-point communication channel (K. Yu *et al.*, 2019).

The architecture of ICN focuses on connecting the consumers to their desired contents. Since the consumer does not bother about the location of content publisher. All the communication occurs in ICN architecture is centered on two main elements. '*Interest*' and '*Data message*'. The interest packet represents the demand of a client for desired content as it holds the name of the content that the requester would like to have. This communication initiates with the names such as content names from the requester. For instance, a user named Bob requested Maradona's Wikipedia page as *org/wikipedia/maradona*. This request contains valid ICN names; as an example, */it/Venice* and */org/Wikipedia/Maradona* are two different valid names composed of different components. In order to fulfil such requests, each ICN node holds the following three data structures:

- ***Content Store (CS):*** CS is basically packet caching that enables an ICN nodes to immediately check if there exists a match with the exact same request to cached data.

- ***Pending Interest Table (PIT):*** PIT maintains the record of unsatisfied requests (interests) that are still in wait for be fulfilled. Also it helps to trace the reserve path towards requester, once the requested content is found.

- ***Forwarding Information Base (FIB):*** The FIB servers as a routing table which is used to forward the interest request towards the node which contains the desired data.

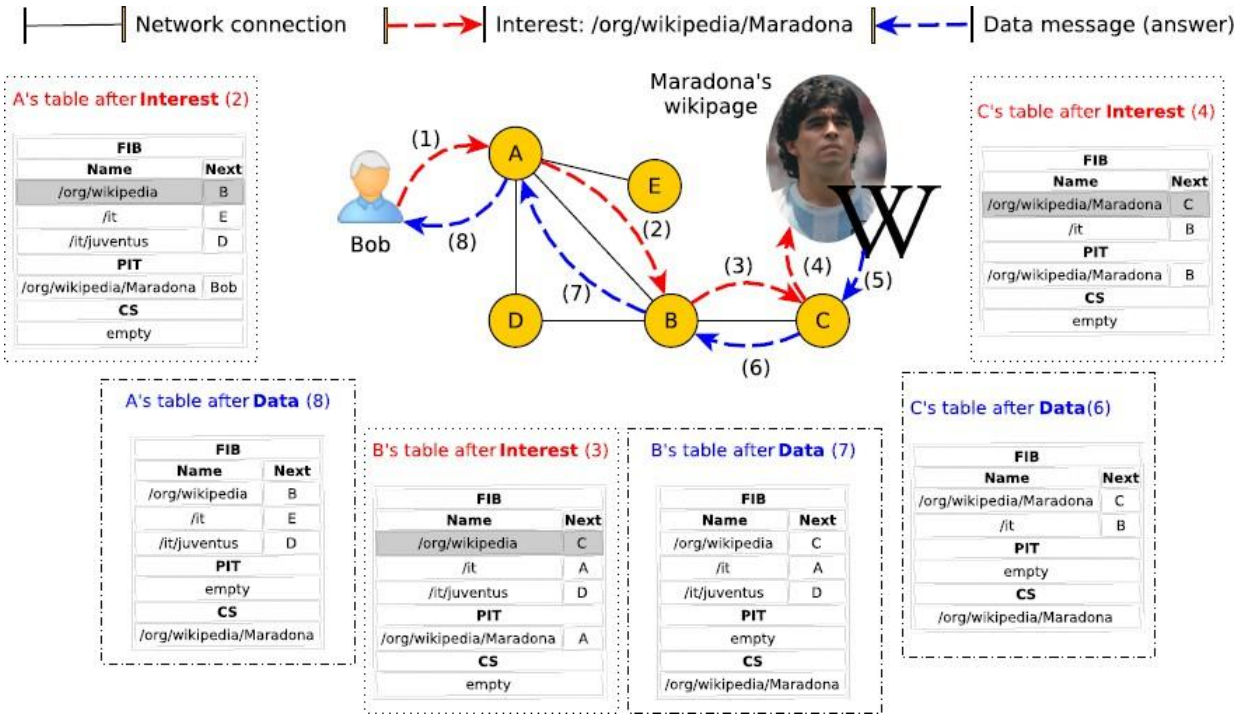


Figure 1-1: ICN Architecture Example

When a client initiates a request (interest packet) and it reaches to the ICN edge node (edge is said because it would be first node connected and hence edge of entire network) the node will then extract the content name from the interest packet and then try to find out in the Content Store if there is any entry exists with same content name. If the entry is found, then the requested content will be sent back to the requester via the same node from where the interest has arrived and then the interest will be discarded. If an entry is found in PIT with the same data, then the packet will not be further forwarded and instead the PIT table is updated with an entry of the node from where the interest was arrived. The content takes reverse path from its origin to ensure the delivery to its original requestor during that the content is replicated to every node.

The figure 1-1 illustrate an example of how content is retrieved and delivered over the ICN network. Status of CS, PIT and FIB is shown after each step. A user named Bob requested a Wikipedia content for a soccer player Maradona. Bob generates the interest as

org/wikipedia/Maradona. Initially the request is received at node A. (i) Node A checks if there is any entry in its CS and PIT with the same data. Considering this example there is no entry found, therefore the request is forwarded to the node B. (ii) apparently a new PIT entry is created indicating interest and the incoming interface of incoming request. After conducting same process as node, A, B forwards the interest packet to C. (iii) as the packet arrives at C, it checks its CS and it finds a match. (iv) The content is then enclosed into data message, (v) which will be delivered through reverse path calculated by entries in PIT table (vi – viii). A copy of the content is replicated on the middle nodes A and B, while the user B eventually receives the requested Wikipedia page (Bernardini and Crispo, 2019).

1.1.1 Motivation of Thesis:

The methodology of ICN works on caching which is in general replication of data that has been requested by. Also, ICN does not relay on point-to-point connectivity either between the client and server or content producer. Due to this reason, the user is not aware from where the data is coming. But this theme of ICN arises several security challenges. For example, as we know that we are only interested in the content and ICN paradigm makes a way to let the user connect to content itself without getting bother from where the data is coming. But there comes a severe need to protect the confidentiality of the data to ensure that none of the unauthorized user can access it. Another aspect of security is to protect the interest of the user, in such a way that the request must be fulfilled without being revealed to any malicious entity and delivering the requested data without any alteration in the content. Several theoretical and practical work has been done in past for the security domain, but attackers are still be able to breach the security. Various surveys have been performed like *A survey of Information-centric networking, A survey of caching mechanism in ICN, Access control mechanisms in Named Data Networks: A comprehensive survey*. But these are not enough to describe attacking strategies that can be executed by an offender and the precautionary steps to prevent them. To the best of my knowledge in this regard, there exists only one survey till dated to document the attacks of ICN. But that is based on the traditional attacks that may occur. This thesis will present a detailed taxonomy for the most recent attacks that could happen in ICN architecture. The main contribution of my work is to provide the research community a detailed approach by explaining all aspects of ICN and presenting attacks and their counter measures in such a way that could be helpful for them to further secure this model from in future security breaches. Furthermore, I am listing some important questions that this work will help to answer:

- What makes the ICN more unique over existing network architecture?
- What kind of security threats could occur in ICN model and how severe those could be?
- Why there is need for new security solutions and how to prevent the most recent attacks?
- What can be the requirements to make ICN more secure?

1.1.2 Structure of Thesis:

This work begins with brief introduction of ICN architecture. Chapter 1 comprised of the detailed scenario of ICN architecture with all of its important parameters. Chapter 2 will describe the detailed taxonomy and classify the attacks that a security offender may carry out in ICN model. The severity of attacks is also been discussed. Section 3 mainly focuses on the counter measures that could be adopted in order to prevent the existing attacks. Chapter 4 explains the access control mechanisms in two categories Encryption based and Encryption independent access control there are already been proposed in ICN.

1.2 Traditional Network vs ICN:

Currently the widely being used network architecture is Content Delivery Networks (CDN). It has been designed to overcome the limitations of existing traditional architecture. The most popular and novel paradigm is ICN. In the year 1999, a named based architecture was introduced by Stanford University of California which was named as Translating Relaying Internet Architecture, integrating Active Directories (TRIAD). This was the time when ICN came into existence and this new model brings the idea of replacing host-based point to point communication with information-based scheme. Such that the user retrieves his desired information by using the content name which being specific location of content publisher. Therefore, at the back of names there exists IP addresses and those addresses are based on the location means that those IP addresses are associated with device that holds a particular location in a network. Changing the physical location which change the addresses as well. With the increases of mobile devices, this will cause huge problems due to constant change of location during specific time interval. The content is being requested by the name itself without being bothered by the location of the host (D. Buitenkamp, 2019).

There are different types of naming schemes have been proposed in ICN but some of the main are mentioned below:

- **Hierarchical names**: These are made up of the set of components that helps to determine the application, also elaborating the service or content. They have similar structure as Uniform Resource Identifiers (URIs). Generally hierarchical names are considered to be user-friendly such way that these convey semantic meaning to user. This scheme also strengthens the network scalability.

- **Flat name**: This type of names is generated by applying hash algorithm to the content. Flat names do not contain any contextual meaning neither a structure.

- **Attribute value based**: As the name states, this type of name contains collection of attributes and each attribute expresses the type, name other possible values regarding content (date of creation, nature of content, version). These attributes report the individual piece of content along with the properties.

- **Hybrid names:** are the combination of the above two naming schemes. The main idea for doing that is to take advantage of the features offered by different schemes in order to keep up the performance and scalability of the network. Also to increase the security and privacy. For instance, utilizing advantage of attribute values based for searching purpose by help of keywords, flat names have the fixed length which is helpful for memory

consumption. Similarly, hierarchical names boost up the look-up process. (Nour and Moungla, 2021)

	Host-centric	ICN
Naming	define host topological position	define content regardless of its location or representation
Routing	between hosts using IP addresses	using name resolution entity or name-based routing
Caching	specific caching points (cache servers)	each node can cache any content passing through it
Security	secure communication channels between hosts	secure the content itself
API	send data to a specific address	publish and subscribe contents

Table 1: HCN vs ICN

1.3 Architectural Components of ICN:

The purpose of ICN is to restructure the existing network model and allow the consumers to access named data objects. While try to achieve these targets by means of ICN paradigm various technical research-based challenges were identified in this model. Several different architectures are also been propose for ICN. But in actual, all the proposed architecture contains some generic concepts. The classification of those architectural components is done below. (K. Yu et al., 2019).

1.3.1 Focus on Information Naming:

Naming the content is an essential component of ICN. For the purpose of massive information distribution the internet has been transformed from being an academic network to a globally existing framework. That includes with over 1 billion devices connections, consisting indexed webpages of around 1 trillion and data been transferred in Exabytes every year (Cisco Visual, 2014). A consumer is now more interested in getting the data/content/information no matter wherever it is located instead of accessing a dedicated sever or content producer. However, despite of the fact that current internet model is based on host-centric communication, while requesting the data, a user must have specify the interested content but also the location of the server where data could be contained. The native layer mechanism of the current internet cannot find and fetch the requested content from the exact location where it has been actually hosted. The community dedicated for researcher and innovations proposed almost decades ago an idea of moving towards information-centric networking from host-centric networks (Gritter and Cheriton, 2001). In the meanwhile, other researches being conducted in the same domain also realized the need of ICN and assuming that it would be operated over the current architecture as an overlay. Similar to the functionality offered in Distributed Hash Tables (DTH) (Mathis ACM, 2002).

The methodology in ICN separates the information from its source and to achieve it clear location-identity split is used as a mean. The main idea behind this scenario is that the content is being named, addressed matched irrespective of its location. Therefore, it can be located anywhere inside the network (Arianfar and Ott, 2010). In ICN, rather the specifying peer-to-peer communication by means of source and destination a content is named. The major advantage of

shifting from host naming model to an information naming model is that the retrieval of requested content becomes receiver operated means that receiver does not hesitate about the fetching the data directly from server.

1.3.2 Focus on Information Delivery:

The general term for this *flash crowds* (which refers to heavy surge over a website during a specific time leveraging a single host with huge number of requests). The current internet model lacks the technique to handle flash crowds, where the data in transit is being dealt by the network nodes in the form of bytes that have to be carried from particular source to a certain destination. These nodes don't have the knowledge of the data they carry and therefore, are unable to perform optimization that be done (e.g: smart caching over the modes, information replication in network, traffic engineering based on information oriented) (Zahariadis, 2011). The ICN network model fulfills the requests for desired information not only by identifying actual information source and also make the use of in-network caching feature that makes possible to contain the copies of desired content (or probably some pieces of data). This is achieved without the need of restoring costly and proprietary add-on solutions (for example Content Delivery Networks to improve performance). As the network layer runs directly above-named information. ICN architecture handles the information fragments in way that they are carrying the named base information. Therefore, data packets can be replicated over the network nodes and can also be retrieved from anywhere within the network very easily. As compared to the current network architecture, where high-cost methods like Deep Packet Inspection (DPI) would be need to use in order the fulfil request with cached data packets over the network nodes (ACM, 2008), (ACM, 2011) keeping in knowledge the fact that DPI does not work with the encrypted data packets. The ICN model also supports accumulation of the requests for the previously requested information, thus enabling its delivery towards the consumer by means of multicast forwarding. At last, the access control (restricting the access for the authorized users only) can be implement on the network layer. The protocols design for delivering data packets to a mobile host imposes "triangular routing" such that the data packet is first sent to its home agent, which is considered as a home network for mobile agent. After that packet reaches to the actual location of host by using a tunnel. This is more ineffective and incompetent way as a packet has to cover longer path than usual. In ICN the host mobility is assisted by publish/subscribe model. When the user needs some information may *subscribe* to it. This will indicate the interest of a client for the information within the network. On the other hand, users *publish* the advertisement with requested information if they are offering the same data that is been requested by a client. Broker are the once inside the network that are responsible for considering a match between subscription and publish. This is a time independent communication between publisher and subscriber. This feature enables the support of mobile user. When the user requesting some data and meanwhile is in mobile state moving from one network to another, it can reissue the request after the network handoff. While the nodes within the network may fulfil the request with replicated data which becomes possible due to caching mechanism in ICN, instead of location original publisher (Eugster and Kermarrec, 2003). However, a challenging situation arises when the publisher itself is in moving state and bringing the question in consideration that how data can be retrieved from a moving publisher (Zhang and Zhang, 2018).

1.3.3 1.3.3 Focus on Mobility:

Existing internet scheme was originated with host situation on a fixed location in a network (IP addresses). The IP address of a host identifies the location where the host is attached. However the observations show the constant increase in mobile hosts accessing the internet (Cisco Visual Networking Index, 2014). Mobile and wireless devices seems to have easy switching between networks in a fashion of changing IP address with respect to their mobility and this introduces new modes of communication. On the other side, the protocols design for delivering data packets to a mobile hosts imposes "triangular routing" such that the data packet is first sent to its home agent, which is considered as a home network for mobile agent. After that packet reaches to the actual location of host by using a tunnel. This is more ineffective and incompetent way as a packet has to cover longer path than usual. In ICN the host mobility is assisted by publish/subscribe model. When the user needs some information may subscribe to it. This will indicate the interest of a client for the information within the network. On the other hand users publish the advertisement with requested information if they are offering the same data that is been requested by a client. Broker are the once inside the network that are responsible for considering a match between subscription and publish. This is a time independent communication between publisher and subscriber. Means that, a publisher may publish the content does not matter if the published data is previously been requested or not similarly a subscriber may request an information after any time the content is being published by a producer. This feature enable the support of mobile user. When the user requesting some data and meanwhile is in mobile state moving from one network to another, it can reissue the request after the network handoff. While the nodes within the network may fulfil the request with replicated data which becomes possible due to caching mechanism in ICN, instead of location original publisher (Eugster and Kermarrec, 2003). However a challenging situation arises when the publisher itself is in moving state and bringing the question in consideration that how data can be retrieved from a moving publisher (Zhang and Zhang, 2018).

1.3.4 Focus on Security:

The Internet was created to be operated under a fully trustworthy circumstance. The requirements like data integrity, authentication, and privacy were not in the count. In other words, we can say that it was designed to be flexible enough to permit new hosts join the network, also it was created to allow any traffic that is being generated into the network by any of the client. However, these unsharpened features of the internet enabled the spammers and security offenders to initiate *Denial of Service* (DoS) attacks within the internet to disrupt the services or network nodes. To tackle down such malicious behavior and security threats, various patching mechanisms and add-on security had been introduced. For example, firewalls, spam-filter also various security protocols (e.g: DNSSec, IPSec). However, all these existing solutions somehow are not enough to coupe-up the security threats to internet and the attackers are still be able to manage their offensive activities in the network (Stuckmann and Zimmermann, 2009). Internet has the certain security issues that are originated due to disconnections between the contextual information at application layer and the data being carried in IP packets. Correspondingly the secure point-to-point connection become dominant, the entire internet model is still not protected from hackers and spammers and the flowing data packets are also not secure (Álvarez *et al.*, 2012).

The ICN network is interest-driven. Means that the data will only flow in the condition when a user certainly request for some piece of information. This will notably helpful in reducing the unwanted amount of data transfer (spam). ICN model focuses on content-based security due to reason that it adopts caching of data over network nodes and also due to naming abstractions. Thus,

securing the content itself is priority (Nour and Moungla, 2021). As the consumer requests a content and information can be retrieved from anywhere inside the network, therefore it is very crucial to make sure that the data is in actual form and its integrity is not compromised. Since the producer/publisher itself is not responsible to authorize the user to access the data various trust models are presented for performing users' authentication (Stuckmann and Zimmermann, 2009), (Sarros and Kapetanidou , 2019).

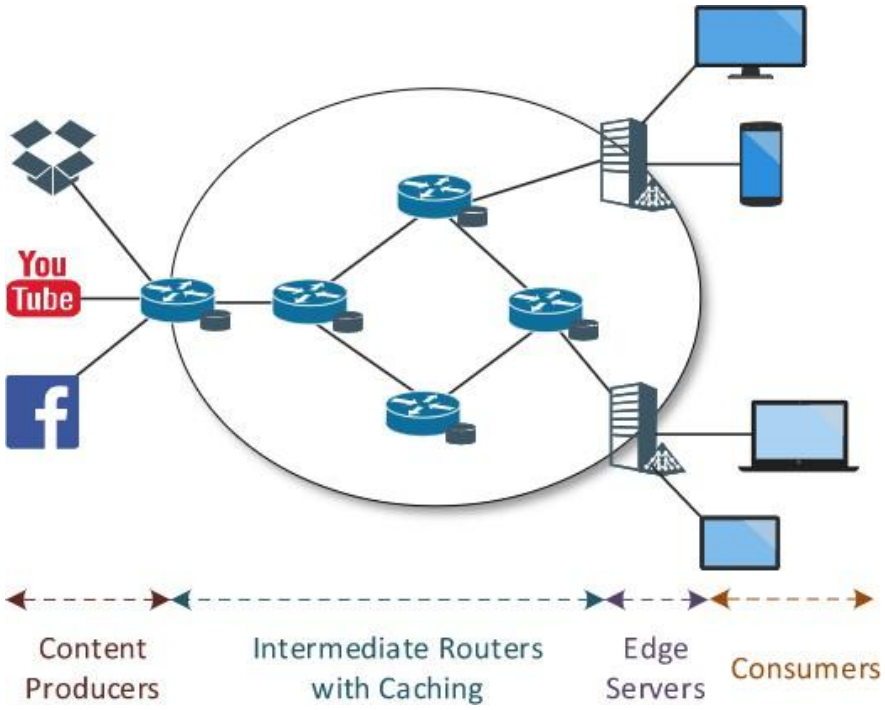


Figure 1-2: Basic ICN Model

1.4 Proposed Information Centric Networking Architectures:

Various ICN architectures have been propose during the TRIAD project. Some of them are discussed below. Note that there are three of architectures that got the highest attention from the community in past and are still indulged as model of choice named DONA, NDN/CCN, PURSUIT. Therefore, particularly these architectures are brief explained below:

1.4.1 Data Oriented Network Architecture (DONA):

DONA was a first ICN architecture that was considered as complete. It was proposed in 2007 by Koponen at UC Berkeley. In DONA model, the hierarchical names are replaced by flat names. As we know URLs are engaged to a particular location translated by their DNS, on the other hand flat name in DONA are diligent even if there is condition of moving information. This enables the availability of information by means of caching and replication over the network nodes. There is a network of *Resolution Handlers* (RHs) servers which are responsible for performing name resolution in DONA. These RHs works in hierarchical pattern in order to publish and retrieve objects.

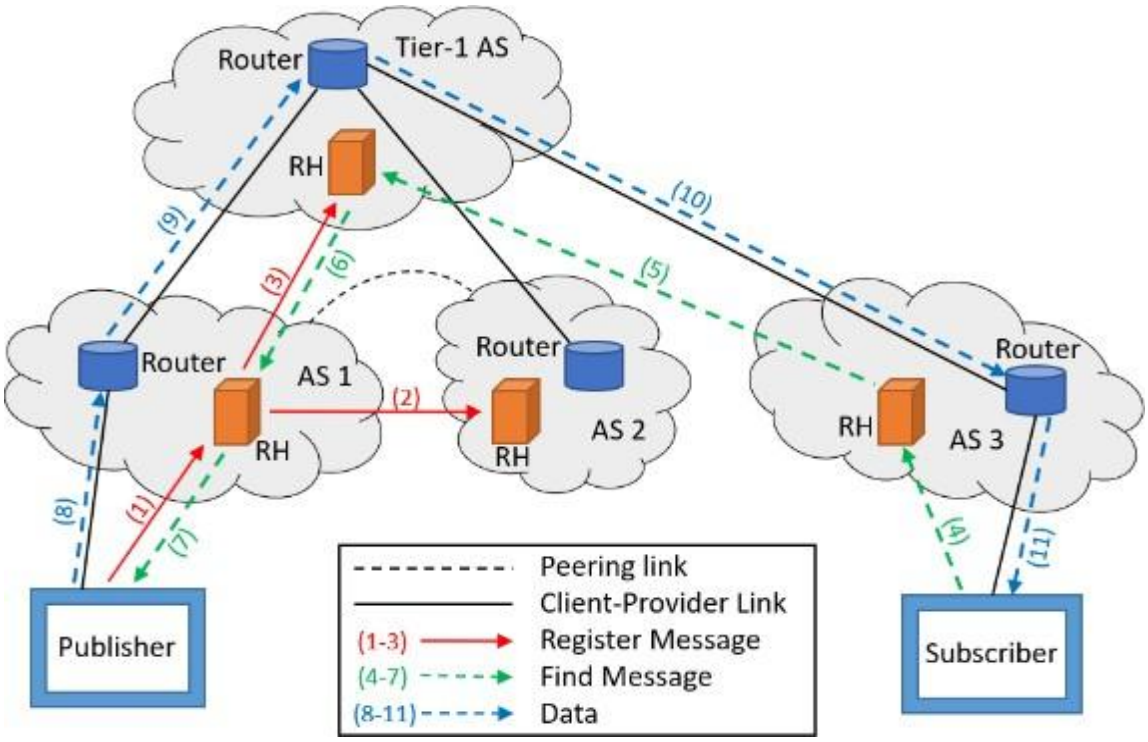


Figure 1-3 DONA Architecture

Illustrating the above diagram, the process starts initially with information publication. For this purpose, a REGISTER message is sent from information publisher towards the RH connected to him (1). Then the RH broadcasts this REGISTER message towards the RHs that are in peering (2-3). This helps to keep a mapping for all the intermediate RHs between name of the object and address of such particular RH that propagates the registration further. This results in replication of registration overall the RHs upwards until tier-1. As every tier-1 provider is peered with together, due to this reason RHs present in tier-1 gained to information of all registration contained in entire network.

In the case of requesting a content, the subscriber/consumer initiates a FIND message towards its local RH (4). This message is further propagated towards the parent hierarchy of RHs on the basis of certain routing policies unless a matching cached entry of the data is found (4-5). As soon as the mapping of data is found, the request is transmitted ahead to let it reach from RH to actual publisher (6-7). From hierarchical point, tier-1 is at the upper most, therefore all the existing objects inside the network in its knowledge. DONA uses two approaches to for data routing namely decoupled & coupled. In case of decoupled, as soon the FIND message is received by respective publisher it sends the data directly to the requester with the help of regular IP routing and forwarding. In the method of coupled, FIND message keeps the track of path-labels while it travels form RH to RH. After the request reaches the content creator, the travel path for the data is retraced back to the subscriber using these path-labels as can be seen in Figure 2 (8-11) (Xylomenos *et al.*, 2014).

1.4.2 Named Data Networking (NDN/CCN):

Content Centric Networking (CCN) was proposed in 2009 at Palo Alto Research Center (Jacobson, 2009). The Named Data Networking follows exactly the same design fundamentals. Therefore, it was selected by National Science Foundation (NSF) in 2010 as a one out of four funded projects by NFS for Future Internet Architecture (FIA). CCN and NDN use the same architectural concepts for instance content caching, hierarchical names and content named based routing. Since NDN/CCN is a client-driven network, a client initiates a request into the network by sending an interest packet (Liebeherr and ACM, 2010) .

In general, which are similar to URLs (for example *programmer.mozilla.org/docs/html*) as discussed previously. However, it is not mandatory that NDN name should be a URL because it does not contain a DNS name as a first part not associated any IP address. Below figure shows a very detailed NDN architecture and elaborates the working.

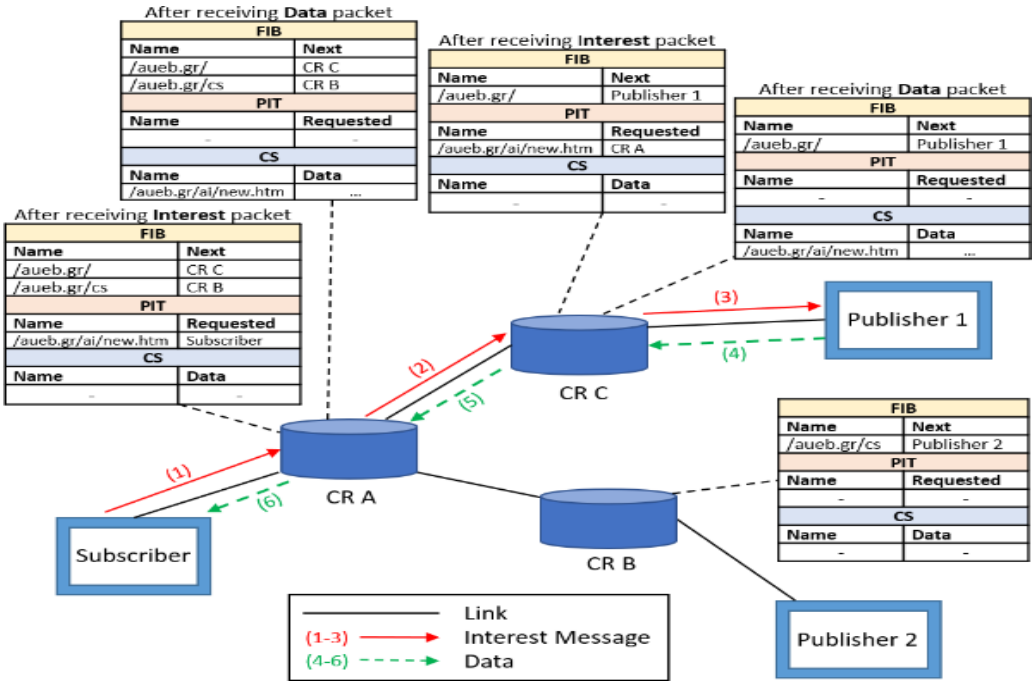


Figure 1-4: NDN Architecture

In NDN architecture, when a subscriber/consumer wants to request a data, he sends an INTEREST packet which carries the name of requested content. While DATA packet is sent is response with the requested information. The transition of Interest message takes place hop-by-hop and reaches to Content Router (CRs). Every CR in the network responsibly stores three data structures Content Store (CS), Pending Interest Table (PIT), Forwarding Information Base (FIB) respectively. The FIB table contains routing information based on names towards output interfaces. So that in future needs, the Interest packet can be directed towards the correct node with respect to the actual content. The PIT table keep the track of nodes from the unresolved Interest packets that has been received so far.

Consider the figure 1-4 as an example. The interest packet */aueb.gr/ai/new.htm* (1) is used as in example here. When the node CR A receives an Interest packet. Initially it extracts the

requested data name and tries to search the content in its local CS and if the name matches with the requested content. In case of no match, the interest message is the propagated to next suitable node which is CR C (2). The node CR C carry out the same steps, now this time the CR C has the direct connection to publisher 1. Therefore, CR C sends the packet to publisher 1 (3). In this way, the requested content has been located so Interest packet needs to be dismissed. The publisher 1 replies with DATA message which contains the requested data (4). CR C then consults its PIT table in order to find out the path to deliver back the content towards its requester. Once the Data packet arrive to every node (5), the CS store is updated accordingly keeping a copy of the data. CR A follows the same process and eventually requester gets the desired data (6) (Fang and Yu, 2018).

1.4.3 Publish Subscriber Internet Technology (PURSUIT):

The Publish Subscriber Internet Technology (PURSUIT) (Särelä and Tarkoma, 2008) is actually forerunner of Publish Internet Routing Paradigm (PSIRP) (Álvarez *et al.*, 2012). Which was a funded project by Framework 7 Program (EU research and innovation program). Under this project, the developed a network model that is clearly helpful for migrating from current IP stack protocol towards Publish- subscribe protocol. The objective of PURSUIT is to provide Internet-scale certain deployable components of the PSIRP model (Bari and Ahmed, 2012). The PURSUIT has three basic architectural elements rendezvous, topology management and forwarding. The rendezvous is responsible for making Rendezvous' Network (RENE) and also referred Rendezvous Nodes (RNs). Topology Managers (TMs) are responsible for managing topology.

The PURSUIT paradigm utilizes flat naming scheme which is very similar to DONA architecture. Thus, the PURSUIT architecture can simply be called as systematic version of DONA model. These flat names are composed of pair of IDs named scope ID and rendezvous ID. The scope ID indicates information objects whereas the rendezvous ID is distinctive ID for selective piece of information. Figure 1-5 depicts a PURSUIT architecture.

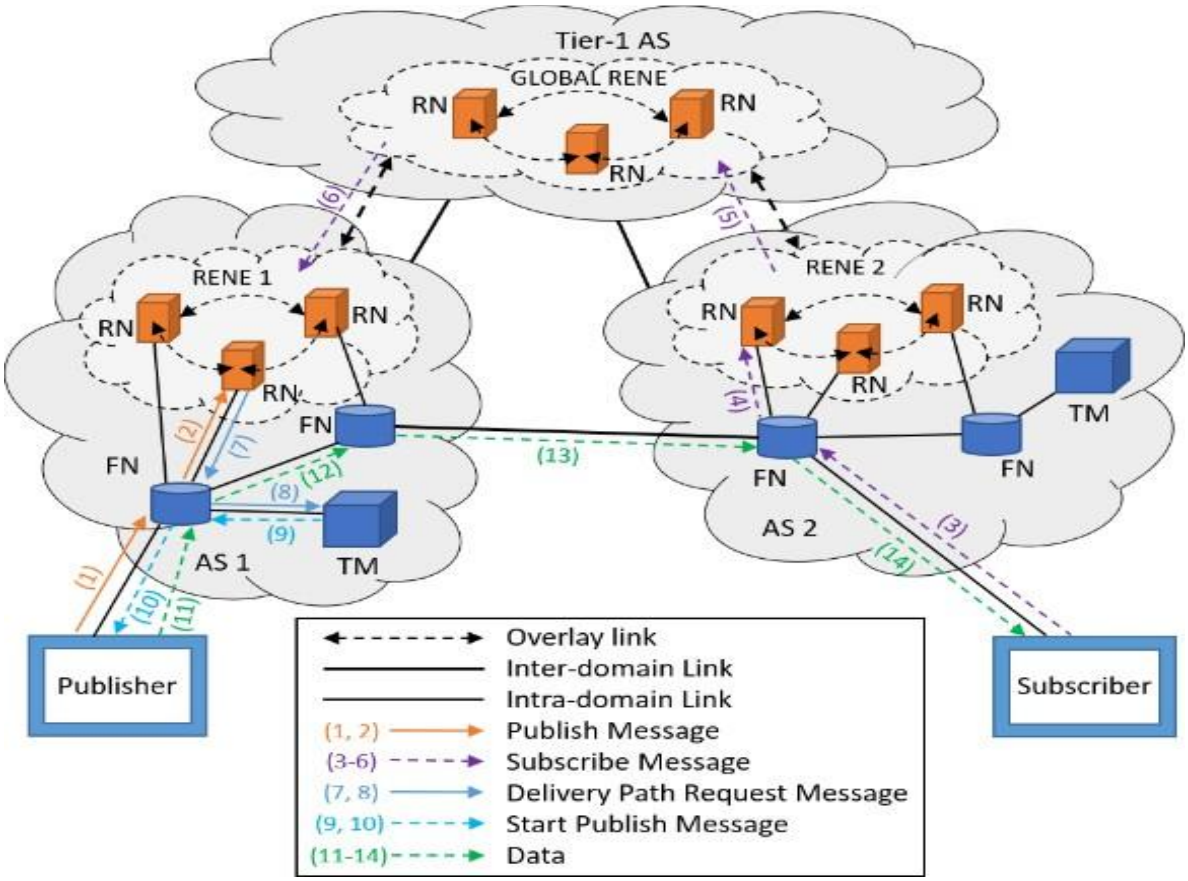


Figure 1-5: PURSUIT Architecture

The rendezvous functions is in charge of handling the name resolution in PURSUIT and the implementation is done by number of rendezvous nodes (RNs) while it uses DHT routing model (Ahmed and Mathieu, 2016). The scenario is very similar to DONA, as the publishers needs to publish an information, it simply generates a PUBLISH message towards locally connected RN and the routing is carried by DHT containing matching scope ID (1-2). On the other hand, if the subscriber needs to retrieve that data, he simply sends SUBSCRIBE message for the same data to its RN. Again the routing decision is made by DHT which directs this message towards RN of publisher (3-6). RN then sends further instructions towards Topology Manager (TM) for the creation of routing path in order connect publisher and subscriber to confirm the content delivery (7-8). The TM uses Bloom filter technique for each flow and appends it with packet header. This route is then sent to the publisher in a fashion of START PUBLISH message (9-10). The publisher uses the same route sent by TM to deliver the content to the subscriber with the help of FNs (11-14) (ACM, 2009).

Chapter 2

Classification of Attacks

The current internet model was established in the past era. During that period, the main purpose of building such design is exchange of packet between two entities or in other words we can say only to organize point-to-point connectivity. While all of this corresponds to connection centric network. The current widely used model is based on IP address which signifies a certain location of a network device. Whereas, the principal objective of ICN is name-based coupling instead of location-based coupling. ICN also tends to revolutionize the client server communication into worthy novel information centric internet (C. Annual, 2018).

There exists different nature of attacks in this paradigm that were never reported previously in the other environments. In addition to that, the previously existing security attacks are also hazardous for ICN model (Igure and Williams, 2008), (Hansman and Hunt, 2004), (Polla and Sgandurra, 2012). The classification of both new and legacy attacks has been performed in this Taxonomy into four categories as shown in figure 2-1. Naming, routing, caching and multifarious respectively. Which can be further categorized depending upon nature. All the attacks are existed impacts the services individually, but there could be a situation that it may have influence on other category as well. For example, a type volumetric attack exists as a type of DDoS attack and other one is unwanted entreaty (request) which is a type of cache polluting attack. They both results in affecting ICN routing and caching. The categorization done in this work is briefly explained below:

Name based Attacks: ICN model has a weak point and face a major warning in terms of privacy because the data requested by the user is revealed in the entire network. Therefore, there is huge probability that an intruder may sneak and take control over the flow of information causing blocking and compromising integrity of the requested content. The attacker can easily block the content flow, by knowing who has requested it also in this way it determines the user's interest (Almeida and Frias, 2011), (Djenouri and Badache, 2005), (Ghodsi and Wilcox, 2011).

Routing Based Attacks: The delivery of data packets in ICN model majorly relies on publications and subscriptions done in asynchronous pattern, this requires additional input to certify the stability in distributed data. Other attacks such as volume based are aimed to disrupt the content exchange and maintenance by impacting memory and processing power that supports these functionalities. The framework if ICN heavily depends on firmness and integrity of data. It is therefore targeted by poisoning names and routes (Wählisch and Vahlenkamp, 2012), (Fotiou and Polyzos, 2010), (Gasti and Zhang, 2013).

Caching Based attacks: Caching mechanism is the one of the major assets of ICN network as it determines the performance of ICN framework driven by receiver aiming to transport the copy of the requested content from the nearest available node. Due to this reason, the ICN architecture is at risk to all the threats that pollute or destroy caching mechanism and all the existing attacks in cache population are aimed to damage ICN nodes caching characteristics (Chen *et al.*, 2013), (Ghali and Uzun, 2014).

Multifarious Attacks: The nature of attacks presented in this category are pursued to misleading ongoing services and gain forbidden access into ICN network. Which may result in inaccurate and uneven spread of flowing data (D. Kutscher, 2012).

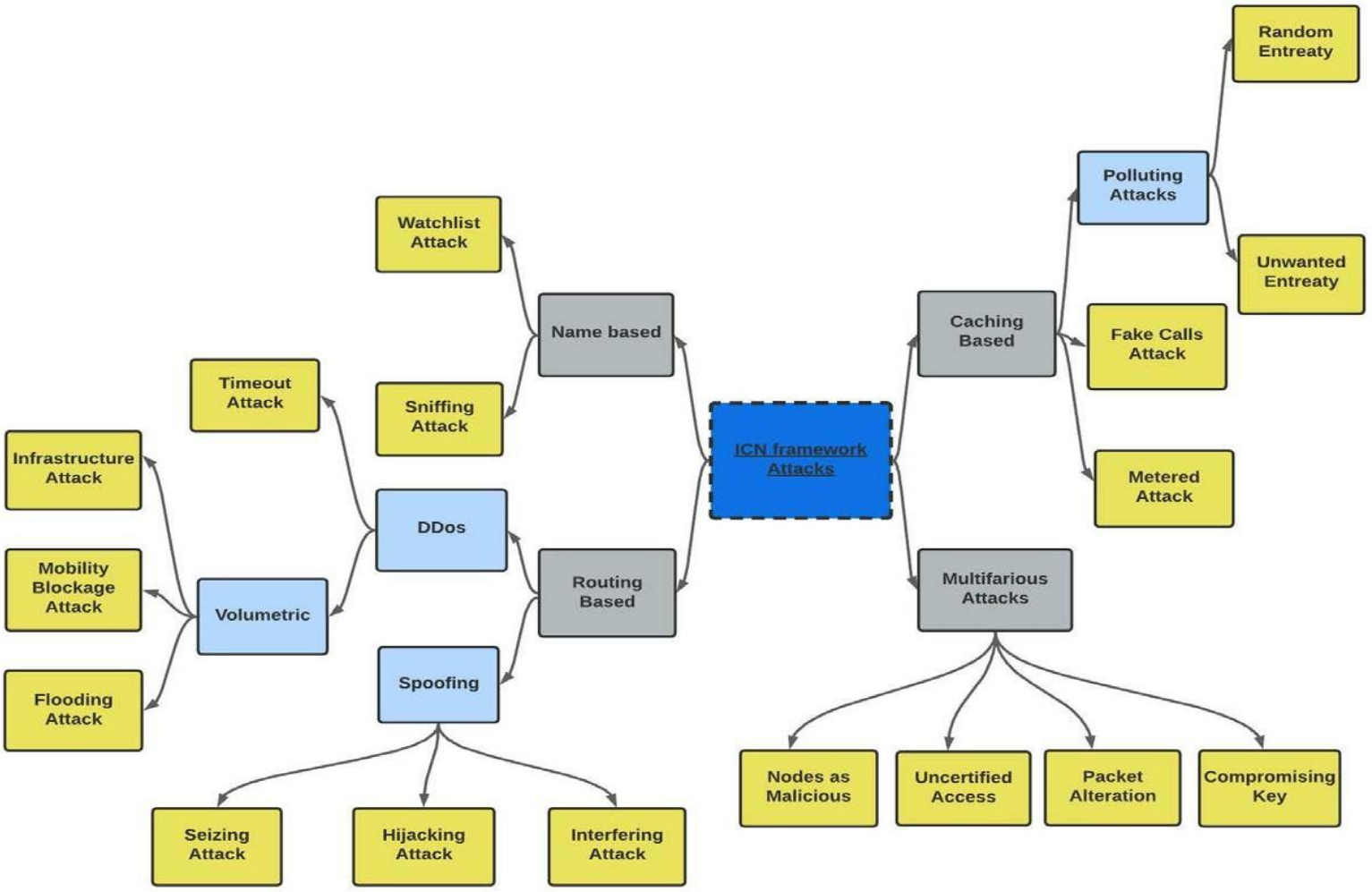


Figure 2-1. Classification of ICN model attacks

2.1 Name Based Attacks:

Name based attacks are further classified into two classes watch-list and sniffing attack. Since we know that nodes in ICN network have overall access to the user's interest. Therefore, the attacker uses this characteristic to launch such kind of security attacks to spoil the system. Taken into assumption that the attacker after breaching the security of an ICN node is able to track and modify ongoing requests (Kutscher, 2011). In ICN paradigm, identification process for the host is not conducted.

2.1.1 Watch List Attack:

The offender contains a predetermined list of requested data with the help of which he can modify of delete the request when matching is done from the list. For this purpose, the attacker performs monitoring of the network so that immediate filtering can be done. As soon as a matching between list and real-time request is found, this gives an opportunity to an attacker to delete the matched request from list or to record user's information. As shown in below figure 2-2, the intruder is sneaking into user's requests to get the knowledge of what kind of data is being requested and by whom.

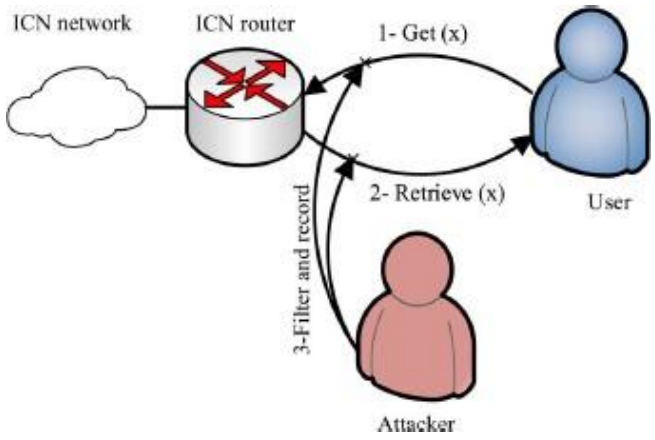


Figure 2-2: Watch list attack: 1- User requested desired content (x). 2- User should get content (x) as per request. 3- With the help of pre-defined list, attacker may perform filtration of requests.

2.1.2 Sniffing Attack:

The sniffing is quite similar to the watch list attack in a way that the attacker sneaks into the system and observes the ongoing traffic inside the network in order to modify or delete it. The content is supposed to be marked by attacker if there exist some specific keywords. Whole scenario of attack is similar to watchlist attack but the only difference present is that attacker does not carry any predefined list to perform matching of data.

Named attacks impact in multiple ways, some of them are discussed below:

- **Censorship:** Carrying out naming attack enables the offender to censor data according to his will.
- **Privacy:** During such attacks, the attacker performs observation of the whole ongoing communication inside network in the form of content requests initiated by requestors. The attacker gets a brief detail about the requesters. ICN model works on user's requests, which is then worsen the privacy state.
- **Denial of Services:** The attacker can stop certain user's requests for the desired data. This leads in having several unattended requests inside the network.

2.2 Routing Based Attacks:

The categorization of routing-based attacks is as follow, Distributed Denial of Services (DDos) attack and spoofing attack. The DDoS can be further classified into volumetric and timeout attacks. Volume based attack has three more classes namely infrastructure, mobility blockage and flooding attacks. Spoofing attack is further classified into seizing, hijacking and interfering attacks.

2.2.1 DDoS Attacks:

Classification of DDoS attack is done as below.

2.2.1.1 Infrastructure:

The attacker initiates bulk number of requests for data that may either available or unavailable. Once ICN router receives these requests, it starts looking for the copy of content from nearby suitable location. These requests are diverted in different routes towards actual source causing congestion inside network. If the amount of these requests if noticeably huge then it leads to denial of service. The severity of attack increases exponentially if in the meantime normal user sends retransmission requests. Regarding the attack, the figure 2-3 illustrates an example. The attacker initiates large number of requests to one of the ICN node with the help of end systems under control. These requests consume huge amount of memory and processing powers while overloading the routing table. As a result, the compromised node forwards these requests to adjacent nodes. If the number of invalid requests from adversary become greater, causing legitimate users to suffer as their actual requests take more time to process.

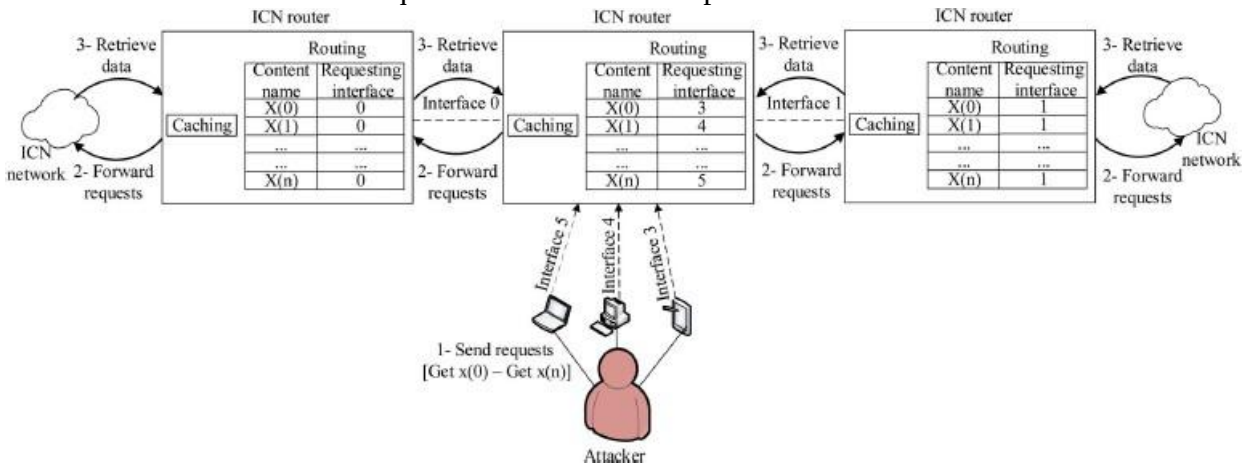


Figure 2-3: Infrastructure attack: 1- Attacker send many request using various end points to a ICN node. 2- The attacked node sends this request to nearby nodes which send further and so on. 3- The network starts getting content for the request from various paths, then forwards it to requested user. These leads to congestion inside the network.

2.2.1.2 Mobility Blockage:

This type of attack occurred when an attacker surcharges the region while passing by through adjacent networks over circular paths and sending higher number of requests. The aim of attacker is to overload all the mobility access nodes to create a state of timeout which later on leads to unavailability of regional networks. The mobility aspect of ICN network model involves retransmission of content request, due to which it is difficult to detect such type of attacks (Bi *et al.*, 2012). A reference picture of mobility blockage attack is presented in figure 2-4. The attack is similar to previously discussed Infrastructure attack; the only difference is that the attacker sends bulk number of requests to the nearest nodes while constantly passing across the networks in circular pattern.

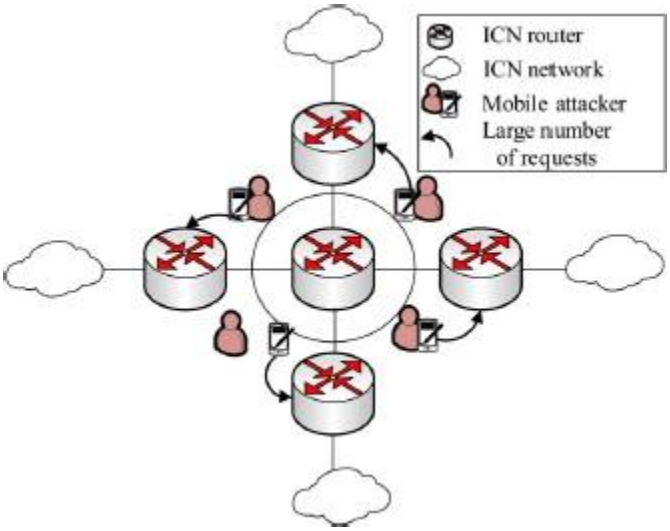


Figure 2-4: Mobility Blockage attack: The attacker is deploying attack while crossing by round the network.

2.2.1.3 Flooding Attack:

The solutions presented to counter flooding attacks in ICN are only be able to limit the number of requests which cannot be considered suitable as a countermeasure for ICN model (Gasti and Uzun, 2013), (You and Haque, 2008), (You and Haque, 2008). An attacker aims to overwhelm the ICN node by sending huge number of bogus requests for non-existing content. Once ICN node receives the requests then forwards it to the adjacent nodes and so on. Once the request reached the publisher, it will drop it since the request is about non-existent content and the entries of PIT will be expired. Attacker sends more new requests. The intent to keep the PIT entries full. This results in packet drop from a legitimate user. The figure 2-5 illustrates an example of interest flooding attack (Vassilakis and Logothetis, 2016).

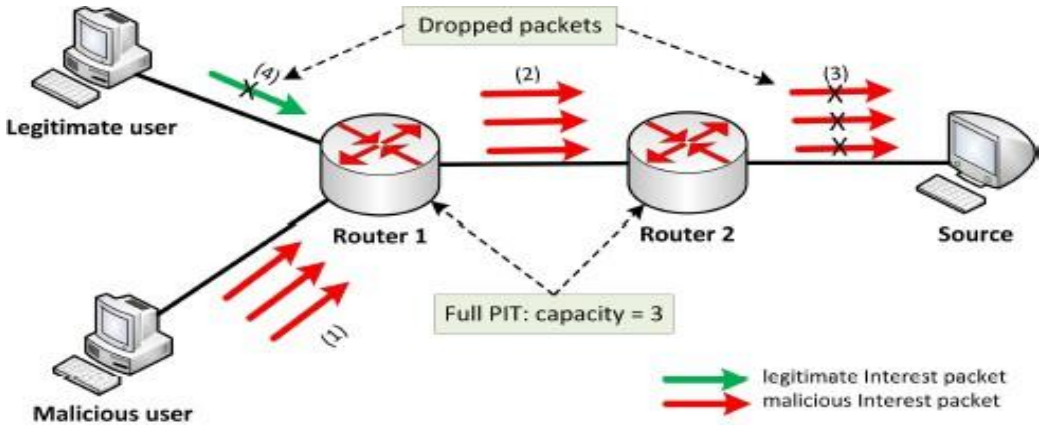


Figure 2-5: Flooding attack: 1- Assuming PIT capacity is 3, the attacker sends 3 bogus requests. 2- ICN node forwards these requests to adjacent router. 3- Source cancels the requests as content doesn't exists. 4- In the meanwhile, request from an authentic user is canceled.

2.2.1.4 Timeout Attack:

The goal of this attack is to increase the request timeout for targeted ICN routers to vandalize the asynchronously organized process of publication and subscription in ICN network. The attacker focuses on degrading execution capabilities of some ICN nodes intent to create longer delays in processing requests and data forwarding. The attack is somehow similar to infrastructure attack because here the attacker is also targeting single router, the only difference is that the offender chooses more than one route to send requests in bulk amount in order to create timeout response for the authorized users. An illustration of this attack is shown below in figure 2-6.

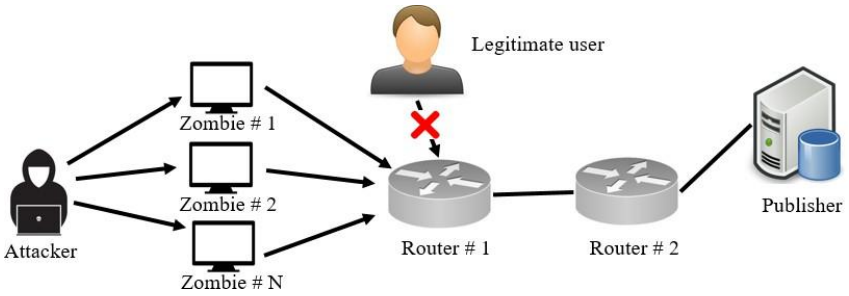


Figure 2-6: Timeout attack: The attacker using multiple paths using malicious devices to attack ICN node.

2.2.2 Spoofing:

Spoofing attacks are classified as below.

2.2.2.1 Seizing:

This nature of attack takes place an attacker starts sending higher number of requests containing maliciously irrelevant content. Initially the attacker disguise himself as a authorize user and a trusted subscriber then initiates these unnecessary content requests to harm to flow of data exchange with in the network.

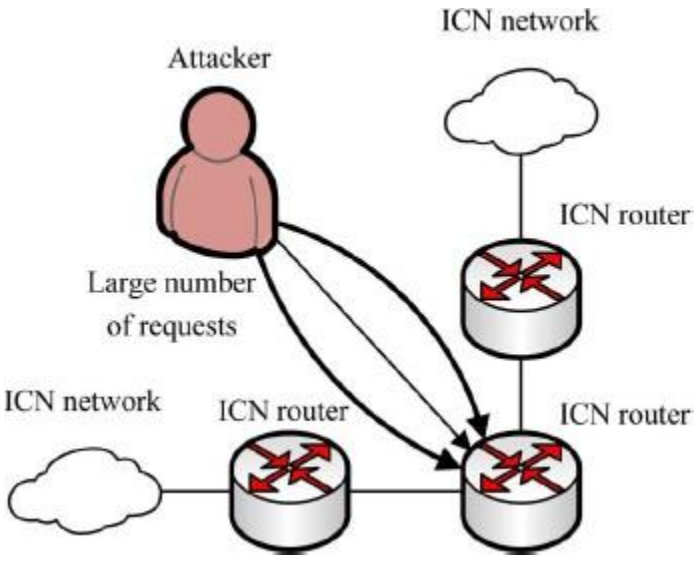


Figure 2-7: Seizing attack: Attacker is initiating bulk amount of requests towards shared router.

2.2.2.2 Hijacking:

Contrasting host centric networking, every node in ICN environment is capable of caching and can do publish/subscribe contents. At first the attacker disguises as an authentic publisher then later may declare invalid routes for the data to disrupt the content delivery. Once any legitimate user within the accessibility of attacker requests for a desired content, this request will go through the fake paths announced by the intruder. Hence DoS will occur. This attack could become worse if the offender extensively hijacks the fake routes. This attack may not harm the system severely because ICN nodes follow the routing schemes that could do attempts into different locations, so the packet maybe not lead to the route faked by an attacker. An example of this attack is depicted in Figure 2-8. Attacker declares fake paths to capture the user's request. When a valid user requests some data, it passes through the malicious paths meanwhile ICN routers further forwards it to the malicious paths in order to deliver content.

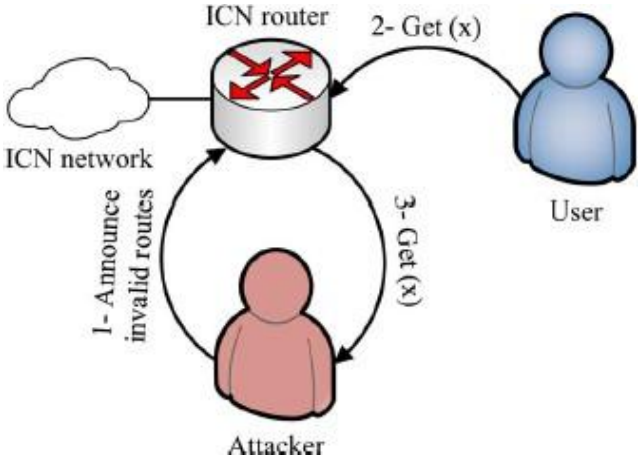


Figure 2-8. Hijacking attack: 1- Routes are faked by an attacker. 2- User sends request for content (x). 3- ICN node took same route to deliver content. Hence content could not be delivered to the user.

2.2.2.3 Interfering:

This nature of attack resembles the man in the middle (MitM) attack. The attacker uses the strategy publishing fake routes by initially disguising himself as an authentic publisher which is quite similar to previously discussed hijacking attack. Attacker captures content requests initiated by users and sends in to correct location respectively. Despite of the fact that user may get the requested data as usual, the attacker obtains the information about the content being requested. Figure 2-9 helps to depict this nature of attack. At first, to capture user's request the attacker publishes fake paths for the content. As soon as any authentic user sends requests for the content, it goes through the malicious path and forwards towards malicious node of attacker by the ICN router. After that attacker forwards the requests further to obtain the requested data. As soon as the data is received at attacker's node, it is then forwarded towards the ICN node from where the request was initially originated and then this node delivers the content to the original requester. The user would never come to know about this whole scenario, while the attacker eavesdrops the whole communication (You and Haque, 2008).

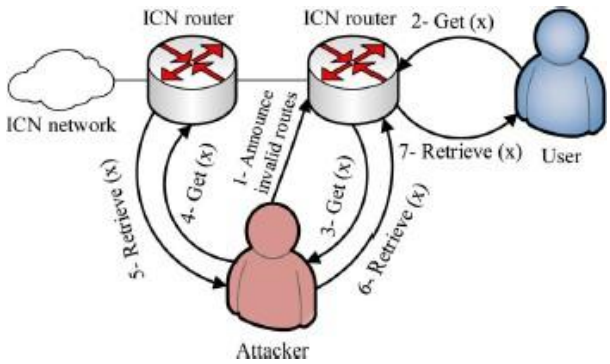


Figure 2-9. Interfering attack: 1- Attacker publishes fake paths for content (x). 2- Valid user requests for same content (x). 3- ICN node forwards request via attacker's route. 4- Attacker redirects this request to get the content. 5- Attacker receives content (x). 6- Attacker sends the content to its original requester. 7- User gets what he requested.

Routing based attacks cause disruption in the following parameters:

- **Denial of services:** This category includes various attacks that can cause denial of services for example initiating bulk amount of bogus request for an invalid content towards an individual ICN node, mobility blockage, hijacking, and flooding and timeout attack. This results in elimination of attack with the exceeding timeout. Which in the long run causes DoS or longer delays at minimal.
- **Resource consumption:** Certain attacks can cause overwhelming the ICN resources. As happens in Infrastructure attack that aimed to disrupt operation by an uncontrolled traffic. For instance, huge number of requests towards a single network entity.
- **Path Infiltration:** As the ICN adopts caching mechanism which includes the distribution of content towards untrusted localities. Therefore, it is typical to validate the content origins as the users is not aware from where the content is being published. Due to this major reason, various attacks like interfering and hijacking takes place that cause path infiltration. In which the attacker at first gains trusts as authentic publisher and then announces fake routes to eaves drop the communication.
- **Privacy:** The privacy is severely compromised in interfering attack in which the attacker gains the access of user's interest. This attack is more impacting when the intruder is in vicinity of the user.

2.3 Caching Based Attacks:

The caching related attacks are categorized into polluting, fake calls and metered attacks. The polluting attack has further division into two more attacks as follow: random entreaty and unwanted entreaty. Detailed discussion of each attack is done as below.

2.3.1 Metered Attack:

This type of attack takes place once an attacker done some analysis of time during the rectification of requested content. As we that every ICN node holds the property of caching any kind of content. The intruder perform calculation on time difference given to response any request for cached and un-cached data. As the figure 2-10 expresses a scenario of metered attack. Assuming that time T1 indicates the duration between which the request for the content is being received and then the publisher provides the requested content at the nearest router. Similarly, T2 is

the time during which request is arrived and content being delivered between user or attacker and the nearest node. If the authenticated user requests a certain content, the ICN network fulfills this request by forwarding it towards content publisher and then returns back to the requestor with the content. Hence the total time elapsed will be T1+T2. If the attacker requests same content, it will be delivered to him within time T2, due to existing cached copy of content in an ICN node. This time difference allows the attacks to perceive if same content is previously requested by the targeted user.

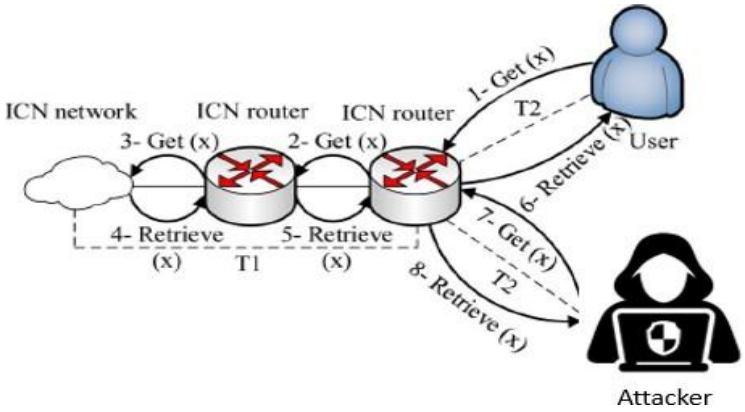


Figure 2-10. Metered Attack: 1- User requests for content (x). 2 -3 ICN network locates content (x). 4 – 5 ICN network delivers back content to the requestor. 6- Time to retrieve content to user is T1+T2. 7- An attacker requests same content (x). 8- The content is delivered to attacker in time T2 due to cached copy at node.

2.3.2 Fake Calls:

In order to violate the caching system, an attacker can send many requests for calling updates for actual content or existing cached copy many times until breaches the convergence time elapsed in routing for fulfilling any content request. This destroys the caching and routing process going on inside the network. As a result, ICN network will no more be able to manage legit requests from authentic users due to this larger number of quick network updates. Overloaded fake calls in the long run led to incomplete or uncertain content delivery. As example of this attack is shown in figure 2- 11.

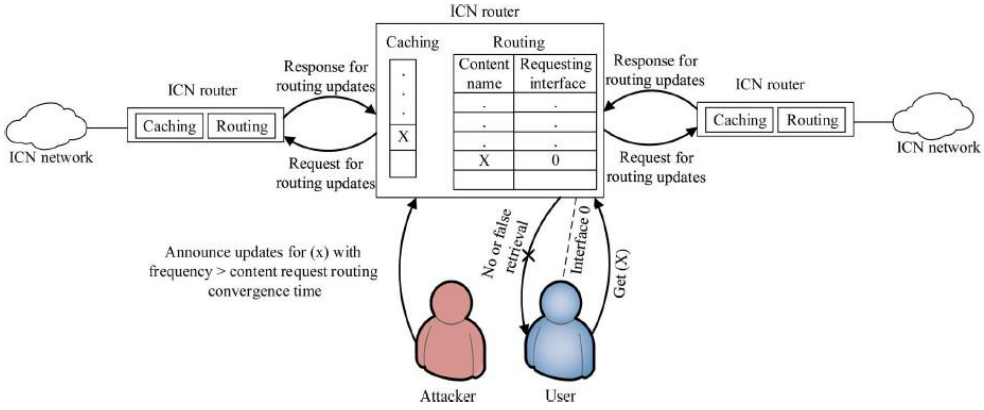


Figure 2-11. Fake calls Attack: User initiates request for content (x). Meanwhile an attacker overloads ICN node with content update requests including (x) in huge number. This bypasses the convergence time. This results in malfunctioning of routing table and then content delivery is unsure.

2.3.3 Polluting Attacks:

Polluting attacks have two types' random entreaty and unwanted entreaty. Each of them is discussed below in detail.

2.3.3.1 Random Entreaty:

This attack refers to destroying ICN caching mechanism and meanwhile modifying content popularity as well. An attacker deliberately requests some unpopular content in order to full the caching storage with copies of such type of content. This unpopular content is the data that is generally not requested by other users. Thus, the attacker initiates multiple requests for this type of content causing the caches full of invalid data. Content can be considered fake if it not requested by authenticated user or someone else modified it other than the actual publisher. Figure 2-12a shows general scenario of data retrieval in ICN network. When a second user initiates request for a particular content, his request is fulfilled by the nearest node because each router in ICN is capable of keeping a copy of content while conveying it towards destination.

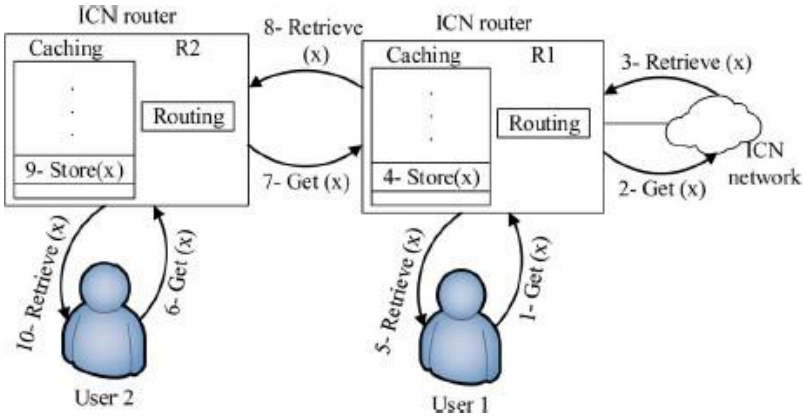


Figure 2-12a: Random Entreaty attack (normal case): 1-User requests for content (x). 2- R1 receives request and tries to locate content. 3- R1 finds the content through ICN network. 4- A copy is kept in caching storage of R1. 5- Content (x) is delivered to the user. 6- Now user 2 requests for content (x). 7- R2 forwards this requests to router to locate a copy as it is already in R1. 8- R2 receives content from R1. 9- Now R2 caches this content. 10- User 2 gets content.

Now the attacked scenario is described with the help of figure 2-12b. The caching mechanism is violated by an attacker by sending huge number of random content requests. After the interval of time if the user 2 requests same data, it takes the whole to fulfill this request same as needed for user 1. Because there would be no existing copy in the cache of nearby router.

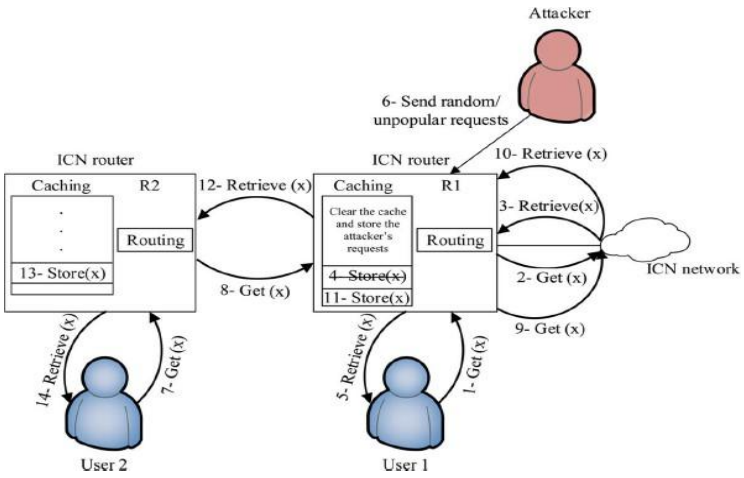


Figure 2-12b. Random Entreaty attack (attack case): 1-User requested content (x). 2-R1 locates content over the network. 3-R1 found the content. 4-R1 kept copy of content. 5-User gets the content. 6-Attacker sends huge number of requests for random data to overload caches. 7- User 2 request same content (x) as previously requested by user 1. 8-R2 forwards request to R1 to find a cached copy. 9- R1 tries to locate copy through network. 10- Content is retrieved from network by R1. 11-R1 keeps a copy of content. 12- Content is sent to R2. 13- R2 also keeps a copy. 14-Finally, user 2 gets the content.

2.3.3.2 Unpopular Entreaty:

This type of attack is initiated by the attacker only with the motive to violate the caching mechanism of ICN network and also to disorder the popularity of content. For such attack, an attacker must have the prior information about the popularity of the content requested over the network. This attack causes same type of disruption as we have seen in random entreaty attack.

All the above discussed caching related attacks harms the following components of the system:

- **Privacy:** Caching system of ICN framework uniformly organized such that every node holds a copy to make sure the content availability. Thus, this higher availability of data leads to have greater privacy risks comparing to current network model. As shown in metered attack, in which the intruder can know about vicinity user that he has previously requested the same data. This information compromises the user's privacy.
- **Cache Pollution:** Anyone including the authorized user is capable of sending large number of requests for random content. This may result in polluting the cache system (Gao and Chen, 2006).
- **Denial of Services:** Fake calls attack leads to certain number of content updates. This causes disruption in data retrieval and once the mapping system is overloaded, there is no chance user can get requested data successfully.

2.4 Multifarious Attack:

This category involves mixed nature of attack. That can be classified as uncertified access, malicious access, packet alternation and compromising key.

2.4.1 Uncertified Access:

Since the copy of requested data is replicated over the entire network. Therefore, it is easy for an attacker to breach the security and get unauthorized access over the data that is being

requested previous by a user or any group of users.

2.4.2 Compromising Key:

This carry is carried out to vandalize the publisher's details extracted from the key and for this purpose any common attack can be used. Such type of attack could have severe impact in ICN framework as the data owners are signing their content that may available over the entire network for a long period. An attacker requests a certain content to like a normal user to breach the publisher's key. Data could be huge in size that it made the attacker's mission of getting publisher's key easier. An example of compromising key attack is depicted in figure 2-13.

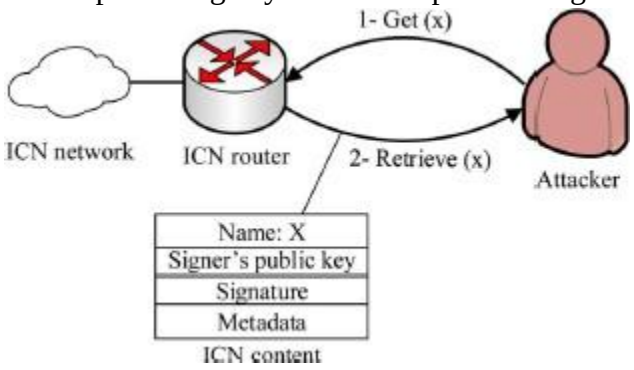


Figure 2-13. Compromising key attack. 1- Attacker requests content (x). 2- Attacker get the requested content which consist of publisher's signature along with public key which later be utilized to extract owner's key.

2.4.3 Packet Alteration:

This type of attack occurs during transmission of data and also consists of replay attack. An attacker somehow deceitfully gets the access over the link and tries to modify the requested interest packet. Even attempts to hinder or reply those packets maliciously certain number of times. During this nature of attack, the attacker carryout following ways: amending of data packets during transmission flow, sending fake replies to the users for their requested content.

2.4.4 Nodes as Malicious:

The ICN framework carryout the distribution of content names as a plaintext. This is huge security threat because any of the ICN node that involves in conveying these data or interest packets can intercept them easily. Once these packets are captured and being evaluated, it will reveal the information about the desired content request from any user. Such harmful or compromised node can also hinder the content delivery towards its requestor just to extract the useful information (Fotiou and Polyzos, 2014).

All of the above discussed multifarious attacks in ICN framework may impact over the following process:

- **Impersonation:** The attacker disguise himself and claims to be either an authentic user or the publisher. The goal is always to disrupt the operation. In case if the attacker successfully gets the key of actual content singer, then he can violate or observe the information and even falsify the communication.
- **Denial of Services:** As can be seen in packet alteration attack, the attacker sends huge number of packets in the response of request or either towards any source. Thus, this

may lead to harm the communication and causes Denial of Service (DoS).

- **Congestion:** The attacker becomes capable of redirecting packets towards already populated paths in the network. This may result in further choking the bandwidth of network links.
- **Compromising confidentiality:** As we know that each network node in ICN has complete access over the interest packets initiated by users for seeking desired content. Therefore, if the hacker breaches the security of any ICN router, he can then easily monitor the requests from specific user. This reveals an attacker about the personal interest of a certain user hence all of the personal information will be at risk.

2.5 ICN Characteristics Impacting Security:

Below are some characteristics of ICN on the basis of which severity of attacks may vary accordingly. With the help of these characteristics, the attacker may deploy his attack with severe significance and that later on become difficult to stop or detect with in the network. The characteristics are as follow:

- **Naming irrespective of location:** This characteristic allow user to fetch the content from any location within the network. Since it could be any location, therefore it can be either trusted or untrusted while user must not be aware of it.
- **State decorrelation:** ICN framework contains two asynchronous natures of states that are request routing and data delivery. There must be firm stability between these two states. In case of failure, it may lead to the situation of DoS or traffic congestions.
- **Caching:** We can say caching is the core component of ICN model. All of the router compiling ICN architecture are capable of caching the data they receive from nearby entity. Later on when the request arrives for the respective content, it can be delivered through certain router without being sent to the actual source.
- **Universal publication & subscription:** All of the users are capable of accessing ICN network from any random location with an identity of either content owner or content consumer. This may also allow some of the user to send unnecessary requests for non-existing content.

With the help of below table, we can relate attacks that can exist in ICN and characteristics of ICN that may cause some sort of impact during these attacks. This table represents how an attacker can take an advantage of these ICN characteristics in order lay down the attack. The letters P and S have been which are acronym for Primary and Secondary respectively. Primary means that an attacker completely relay on this characteristic to deploy his attack and without this characteristic the attack may not be possible. Secondary means that to some extent attacker is relaying on this characteristic or maybe to boost up the disruption during attack.

Attack	Naming irrespective of location	State decorrelation	Caching	Universal publication & subscription
Wathlist attack	P			
Sniffing attack	P			
Infrastructure attack		P	S	P
Mobility blockage attack		S	S	P
Flooding attack		S	S	P
Timeout attack		S	S	P
Seizing attack		S	S	P
Hijacking attack	P		S	P
Interfering attack	P		S	P
Metered attack			P	
Fake calls attack	S	P	P	P
Random entreaty		S	P	P
Unwanted entreaty		S	P	P
Uncertified access	S		P	S
Compromising key	S	P	S	S
Packet alteration		S	S	S
Nodes as malicious	S		P	P

Table 2: ICN characteristics impacting security. (P- Primary; S- Secondary; blank; doesn't affect)

2.6 Severity of Attacks Caused in ICN:

The severity chart for the evaluation of each ICN attack based on ten standards is shown as below. Some of these standards are based on ICN architecture. These standards are used to measure the severity of attack. The three classes (High, Medium and Low) have been used to identify the threat level of an attack based on the following standards.

- **Content Blocking:** H: All content is blocked of each ICN user. M: Blocks content only for users in neighboring paths. L: Only blocks for nearby users.
- **Access over user's requests:** H: Can access any request from any ICN user. M: Can access only requests from nearby users. L: Only requests from certain users can be accessed.
- **Polluting cache:** H: Damages every cache. M: Huge number of caches are damaged. L: only few caches are affected.
- **Spreading attack:** H: Attack widely spread without constraints. M: Some locations are under attack. L: Only a little privilege got by an attacker.
- **Misrouting.**
- **Time-out.**
- **Affecting nodes.**
- **Attacked network distribution.**
- **Remote misuse.**

Above 5 standards have standards has affect as follow: H: attacker affects and dominated large area of ICN network. M: Only neighboring networking entities are affected. L: Only specific nodes were under attack.

- **Restoration:** The restoration after an attack is directly proportional to the previously mentioned standards misrouting, affecting nodes, spreading attack, remote misuse and attacked network distribution.

The table 3 shows the severity caused by each ICN attack. These attacks are classifying into high, medium and low respectively.

Attacks	Blocking content	Access over user's requests	Polluting cache	Misrouting	Time-out	Affecting nodes	Attacked network distribution	Remote misuse	Spreading attack	Restoration
Wathlist attack	H	H			L	L		L	H	H
Sniffing attack	H	H			L	L		L	H	H
Infrastructure attack	H		H	H	H	H	M	H	H	H
Mobility blockage attack	L		L	L	L	L		M	L	L
Flooding attack	M		M	M	L	L	L	M	M	M
Timeout attack	M		M	M	L	L	L	M	M	M
Seizing attack	L		L	L	L	L		M	L	L
Hijacking attack	L		L		L	L		L	L	L
Interfering attack		M	L		L	L		L	L	H
Metered attack		M			L	L		L	L	H
Fake calls attack	H		H	H	H	H	H	H	H	H
Random entreaty			M	M	L	M	H	M	M	M
Unwanted entreaty			L	L	L	L	H	L	L	L
Uncertified access		M			M	L		H	H	M
Compromising key					L			H		H
Packet alteration	L		M	M	M	M	L	M	M	M
Nodes as malicious	H	H	L			H	M		L	M

Table 3: Severity of ICN attacks. (H. High, M. Medium, L. Low)

Chapter 3

Counter Measures against Attacks (Mitigation)

ICN architecture involves various number of privacy and security risks as comparing to the currently deployed internet model. That is all because of the default working model followed in ICN paradigm. There are four security conditions namely integrity, confidentiality, availability and privacy that must be fulfilled by a solution be considered as an effective against this attack. Integrity means the solution must be able to detect any changes that are made in the content and corresponding metadata either deliberately or accidentally. Confidentiality means that only authorized individual can only have access to the secured content. Availability confirms that the information being published within the network should only be available for the authenticated users. Privacy constitutes the protection of both the user and the information. The table 4 indicates the correspondence between ICN attacks vs security requirements. This table is formulated with the help of OWASP risk rating.

Integrity: (alteration of data). H entire data is impacted. M large amount of data is impacted. L only small amount of data has an impact.

Confidentiality: (data compromising). Hall of the data is compromised. M great amount of data is disclosed. L only few data is affected.

Availability: (services deprivation). Hall on going services have been impacted. M considerably high services have an effect. L only some services have impact.

Privacy: (disclosure of secret information). H every user has been revealed. M users in vicinity are affected. L only targeted users are impacted.

Attacks	Integrity	Confidentiality	Availability	Privacy
Wathlist attack		L	L	H
Sniffing attack		L	L	H
Infrastructure attack			H	
Mobility blockage attack			L	
Flooding attack			M	
Timeout attack			M	
Seizing attack			L	
Hijacking attack		L	L	
Interfering attack		L		M
Metered attack				M
Fake calls attack			H	
Random entreaty			M	
Unwanted entreaty			L	
Uncertified access		M		
Compromising key	H	H		H
Packet alteration	H		M	

Nodes as malicious	M			H

Table 4: Security requirements of ICN attacks.

3.1 Security solutions against ICN attacks:

Many solutions have been presented to counter the security attacks in ICN. However, these solutions are still having some loop holes. To some extent these solutions may match the point to provide defense against existing attacks but still concrete mechanism for ICN security is still a question to be resolved. Below is detailed discussion of security present yet to mitigate the ICN attacks. The solutions are categorized based on the classes of attacks.

3.1.1 Defense against Name based attacks:

Some solutions have been presented to tackle down the name-based attack irrespective of the model being used such as onion routing (TOR) (Dingledine and Syverson, 2004), untraceable digital pseudonyms (D. L. Chaum, 1981), freedom 2.0 (Back, Shostack, 2001), free-net (Clarke and Wiley, 2002) and deniable encryption (Canetti and Ostrovsky, 1997) are not suitable for ICN model as all of the proposed solutions are not supported for the framework where content is cornerstone. These solutions are applicable with the having specific storage device, pre-shared information between publisher and the requester. The ICN architecture requires solution that can ensure censorship, isolation and credible denial-ability for the users. The solution should also be user friendly such that the requester may not face any trouble in retrieving requested data.

On preserving privacy in Content-Oriented Networks; (Arianfar and Shenker, 2011) in this work, the authors presented an approach that is independent of exchanging any kind of public or private keys between owner and requester of the content. This work involve improvement in the content-oriented framework. However, the authors are well aware of the fact that this work does not provide an ideal or solid security mechanism for the ICN architecture as their work is based on several sort of assumptions

that may not be applicable over ICN framework also this solution is suitable when there are large number of users.

Toward Content-Centric Privacy in ICN; Attribute based encryption and routing; (Ion and Schooler, 2013) this solution includes a proposal a scheme for ICN privacy with the aim of strengthening data confidentiality. An attribute-based routing model was developed to secure the interest of requester. The motivation to propose this model came from the application layer public-subscribe scenario. The idea involves soliciting the access control policies to the content.

Named data security scheme for the named data networking; (Hamdane and G. el Fatmi, 2013) in this proposal an enhanced security mechanism was present for ICN/NDN/CCN architectures. Initially they suggested security requirements for naming the content then the main idea includes a scheme by merging the two cryptographic systems namely Public Key Infrastructure (PKI) along with the Hierarchical Identity-Based Cryptography (HIBC). A secure method of naming the content is still a question mark in the ICN architecture that still demands an efficient resolution. As a suggestion it could be favorable if a naming scheme includes metadata. For example, calculating hash of a content and publisher's signature and identification to strengthen the security. Such as, a secured content naming mechanism can be considered as signature of manifest with chunk names

and hash by the content owner.

3.1.2 Defense against Routing based attacks:

Several numbers of mechanisms were presented to mitigate the routing-based attacks particle to counter down the DDos attack. Some of the solutions are based on rate limiting which is nearly impossible in ICN to do as the framework of ICN does not have host identifier. Taking the advantage of the loopholes, the attacker can easily send large number of requests which can exceed the threshold of requests being handled in per unit time.

Prevention of DoS Attacks in Information-Centric Networking; (AINS. IEEE, 2018) various solutions are presented by authors to counter the different nature of DoS attacks for different architectures of ICN. In order to counter the Interest Flooding Attack (IFA), initially the authors highlighted the problem faced during the defense if IFA that is detect the anomaly in the interest packet. Therefore, authors presented an idea of calculating rate of PIT expiration for every single named prefix and also set a threshold to spot the malicious prefix.

Preventing DDoS attacks in Path Identifiers-Based Information Centric Networks; (Al-Duwairi and Ozkasap, 2020) to counter the DDoS attacks a hybrid system model was presented named GET Message logging-based filtering (GMLF). However, the general NDN architecture follows Pending Interest Table based routing mechanism for content delivery. In Path Identifiers (PID) based ICNs, the paths to carry the requests between network nodes are added in the interest messages. Such that the content can be delivered back to its requester using the same path. Their proposed idea involves Bloom filters-based logging which is used to store GET messages.

DoS flooding attack in Information Centric Networking; (T. Rao and K. Kumar, 2018) in this paper, the authors simply adopted solution to detect and block the DoS attack by using the Snort software. The authors deploy a snort-based rule by setting a threshold to monitor the traffic. This solution is only applicable for the single network because the policy created in snort cannot handle a large number of users also as in ICN every host can either be a requestor or the content producer.

Interest flooding attack and countermeasures in Named Data Networking; (Afanasyev and Zhang, 2013) in this approach, three different algorithms were presented to mitigate the interest flooding attack DDos attack. The algorithms include: Token bucket with per interface fairness, satisfaction-based Interest acceptance and satisfaction-based push back. The authors performed simulation for each algorithm and the results shows that every algorithm is capable of shutting down the attacker while the Interest messages from authenticated users are being processed appropriately. After conducting the assessment of all the three algorithms, the authors indicated the best algorithm which is satisfaction-based pushback among all the three others. As this algorithm shows the best performance in mitigating the DDoS interest flooding attack with minimal effect over the legitimate users.

A defense mechanism against Timing Attacks on user privacy in ICN; (Sivaraman and Sikdar, 2021) Newest and most recent proposed security solution to counter the timing analysis attack. In this paper, the authors presented a distributed caching mechanism for content. The scheme is consisted of two metrics differential privacy and mutual information from information theory. The presented solution provides optimal results in comparison to the other existing solutions also a suitable

caching structure for ICN nodes.

Several other solutions are also present as counter measure for DoS attacks but those are not best fit for ICN model as they can be useful in IP based networks. All the above discussed solutions for routing-based attacks particularly for mitigating DDoS attacks are suitable enough and can be adopted in ICN. But so far these strategies are either implemented on a particular ICN architecture or can only be suitable to counter specific nature of attack in DDoS (for example interest flooding, timing analysis etc).

3.1.3 Defense against Caching based attacks:

Caching based attacks are applicable to all ICN frameworks. However, could be less harmful in a framework which uses self-certifying names. In the self-certifying names, the packet name is within the digest of packet's content. Therefore, it becomes simple to substantiate the actuality of the content chunk. Calculating the hash between content chunk and the digest and discard the packets which do not satisfy the hash. Several security solutions are present to safeguard the caching mechanism from being breached, but yet now none is satisfactory to match the security of cached data.

Protecting access privacy of cached content in Information Centric Networks; (Chen *et al.*, 2013) the authors in this paper initially demonstrated the timing analysis attack over caching mechanism in ICN architectures. Then presented solution for timing attack by introducing three different protocols. The parameters for these protocols contained the time delay to responses given by an ICN node to fulfil a certain content request.

Mitigating content poisoning attack in Named Data Networking; (Ghali and Uzun, 2014)] in this work the author presented an approach to mitigate the content poisoning attack and formalized the meaning of fake content. As per the authors, the fake content is said to the one having valid signature but the wrong key. A scheme is proposed as ranking algorithm for cached data centered on exclusion-based feedback. Exclusion is basically a characteristic of NDN and CCN frameworks that allow a user to exclude a certain content from matching his interest (Mosko and Wood, 2019). The ranking is totally based on user's action upon receiving the invalid data. All the NDN nodes constantly gather statics about the content rejected by the consumers and allocates a numeric value to every cached data. This ranking is assigned on the basis of three strands; exclusion time, exclusion interface ratio and number of exclusions.

Elements of trust in Named-Data Networking; Network layer trust in Named Data Networking; (Ghali and Uzun, 2014), (Ghali and Uzun, 2014) the authors of previously discussed solution deeply analyzed the cause of content poisoning attack in ICN model. According to them the mitigation for content poisoning is conditional over the network layer trust regulation. Conforming to the authors, the root cause of cache poisoning attack is discrepancy and absence of trust model. The attack occurs due to couple of reasons. First, due formation of interest packet as it only relies on content name leaving behind content digest and publisher public key digest (PPKD). Second is due to inexistence of trust model over network layer. To overcome these deficits, the authors presented a solution to remove the uncertainty of interest packet by creating a mechanism named "Interest Key binding (IKB)" carried

out at the core routers. The process involves binding between public key of publisher and the

content name. This scheme is achieved by adding public key of content publisher with the content's key-locater. One of the weak assumptions while presenting this solution that authors have made is that they have considered the core routers as the trusted ones. In case if the router is compromised, false Interest Key Binding will be carried out. Another downside of this solution is that conducting signature verification with Public Key Infrastructure (PKI) is very slow and cannot perform well at line speed.

A lightweight mechanism for detection of cache pollution attacks in NDN; (Conti and Teoli, 2013) in this work a machine learning based solution was presented to mitigate the cache pollution attacks. Initially the authors simulate the cache pollution attack to show that how it can cause the disruption into the system irrespective of the network topology. The authors claims that proactive countermeasures are not productive when the adversary is realistic. Then a detection algorithm is proposed that works over sub-routine of caching policy. The composition of this algorithm is done in two steps; learning and attack-testing.

To counter the content poisoning attacks, so far it has been understood that the approach based on hash verification has provided better results and considered to be more effective. The only requirement is to found an appropriate hash function. Another method to counter such attack could be to detect the source from where the fake content is being originated by leveraging details of interfaces over nodes. For example, a router may be prevented from caching data from an interface that has been declared as suspicious or contains same title as a fake content. Such an attempt will help to reduce fake calls attacks.

Another key aspect for mitigation is designing the strong mechanism for caching. Which can guarantee the user's experience and also improves the network latency while providing significant durability against caching attacks. One approach to consider is exploring in detail the collaborative caching strategies. The collaborative schemes and exchange of feedbacks between caching nodes can help to eliminate the cache pollution attack. For example, an alliance between collaborative caches can exchange information about caches that will help to prevent caching unpopular data (Saha and Yla-jaaski, 2013), (Mick and Misra, 2016).

3.1.4 Defense against Multifarious attacks:

Not all the solutions are applicable in ICN architecture that are presented to counter the multifarious attacks. The reason is that ICN follows a replication mechanism due to which the several copies of data are available within the network including the caching routers and servers.

Access Control enforcement delegation for ICN architectures; (D. Kutscher, 2012) in this work the authors provide the solution for unauthorized access attacks in ICN architecture. The scheme presented as a solution allows the publisher of an information to validate the request against the access control policy without accessing the credentials or any data of the actual user. The system design consists of four entities, data owner (Owner), consumers (Consumer), replying party (RP) and access control provider (ACP). The owners are actually the publishers of any content and have a full control over it. Consumers are the requestors of the data those are interested in certain piece of information. The relying parties can be considered as replication servers that store the content and are also responsible for systematic distribution of content from storage to the legitimate requestor. While access control provider (ACP) acts as an LDAP.

Detection of malicious nods in NDN VANET; (Zhou and Dong, 2017) a mitigation strategy was proposed in this work while detecting the malicious nodes with in the network when NDN is deployed into Vehicular Ad-hoc Network (VANET). The attack that occurs in VANET due to malicious nodes is named as Interest Packet Popple Broadcast Diffusion Attack (PBDA). This malicious node could take dominance of nodes mobility and fabricate fake Interest packets and send them with greater request rate comparing to the normal nodes. The purpose of this attack is to overload the PIT table. The authors claims that there is no solution presented until now to counter this security attack and thus present a strategy based on reputation value (RV) to detect malicious nodes. Every router within the network computes RV for adjacent RVs and uses Markov chain for predicting current RV state based on its historical RV results. Now the resultant RV is used to make decision either to reject interest packet or not.

There are some other solutions presented for smart grid data collection (Kim and Thottan, 2012), (Kim and Thottan, 2011) that can be studied and checked their compatibility for ICN framework. Smart grid is used to manage large amount of data produced by measuring devices. Furthermore, solutions to counter the web application attacks are presented (Barua and Zulkernine, 2011), (Shahriar and Zulkernine, 2010).

The above-mentioned solution for smart gird and web application attacks can also be adopted for ICN architectures with some adjustments to make it suitable to leverage with the location independent content-based caching mechanism. It is not feasible to rely on the shared keys as every user is capable of publishing and subscribing any content. ICN model is also independent of standard client-server authentication procedure. Therefore, adding any authentication server will harm the rigidness of architecture. Although security is a major aspect to deal in ICN. Several researches have been done and still going on to protect the model from attacks. But these are either depicted to counter one of the attack or for a specific ICN framework.

Chapter 4

Access Control Mechanisms in ICN

Access control (AC) mechanism is the backbone for today's internet architecture. AC policies ensure the authorization and determine only the certain type of data must be accessed by the authorized user(s) (H. C. A. van Tilborg, 2005). The current TCP/IP based networking adopts access control mechanism that is followed between client and server, because the server is responsible to conduct the authorization process and permitting the user to access the content. In ICN, all the communication is centered on the content name and thus the content is retrieved from its origin towards to requester. This new means of satisfying requests brings certain challenges to put in consideration as the control over the content is lost. Genuine attempts have been made by presenting different solution to counter this issue. But also, there exists some solution that takes back the authentication towards actual producer enforcing user to get the authorization from the content creator in order to access the data. Although, such solutions miss the uniqueness of in network caching and faces failure due to content publisher's absence (Y. Yu UCLA, 2015). The most active and popular ICN model is Named-Data Networking (NDN) which provides clear concise perspective for Network communication. In NDN the focus is on securing the content itself rather than securing communication channel unlike in the TCP/IP based network scenario (H. C. A. van Tilborg, 2005), (Fang and R. Yu, 2018).

Classification of Access Control mechanism is done into two categories, Encryption-based Access Control and Encryption-independent Access Control respectively. However, there are further subcategories for each. Various proposed access control mechanism for the existing IP based network as well as their pros and cons are discussed below.

Identity-based encryption: it is based on Public Key Infrastructure (PKI). It involves a private key generator, which is responsible for generating public & private key respectively. As soon as user satisfies the authentication process by providing credentials to the key generator. User may then be permitted to access the content. The goal of this method is to minimize the complexity of encryption process. But indeed, the whole model risks at having faith on the 3rd party which is performing authentication for user and generating master key.

Attributes-based Encryption: In this method, the content producer receives public and private from an external attribute authority. The owner permits the user to content by conducting authorization based on certain attributes also produces keys that are eventually used for content encryption. This method leads to friable access control; attribute authority uses public key to generate key and content encryption for each user.

Key-based Authentication: In this scheme, the encrypted content contains attributes set. A third party issues a private key which is correlated with the access policy structure that identifies the user. Only in the case when user's private key access policy matches the attributes that are existed with the content, a user may then be able to access the information. This provides firm access

control based on attributes attached with content. But the use of additional access control structure produces overhead.

Multi-authority Authentication: As the name says, it involves multiple authorities that are providing access policy structure based to given attributes of the user. The content publisher performs the encryption with the help of symmetric encryption techniques also divides it into multiple chunks before encryption. Only by satisfying access policy related to the content, a user may than access the information. However, insertion of various attributes and authorities could leads to some serious overhead (Nour and Moungla, 2021).

Approach	Features	Level	NDN Applicable	Storage	Computation	Drawbacks
Identity-based Encryption	• Generates public keys from known unique identifiers • Does not require distribution public keys	C, P	✗	M	L	• Third-party authentication • Requires a centralized server • Requires a secure channel
Attribute-based Encryption	• Uses a set of attributes • Provides fine-grained access control	C, P	✓	L	L	• Challenge of attribute revocation
Role-based Authentication	• Use of the role concept • One-to-many communication model • Key revocation	P	✗	H	H	• Additional overhead • Not suitable for dynamic network conditions • Third-party authentication
Key-policy-based Authentication	• Provides fine-grained access control • Determines which attributes to use with content	C, P	✓	H	H	• Additional overhead • Third-party authentication
Multi-authority Authentication	• Uses attributes from different authorities • Uses symmetric encryption • Expressive and efficient scheme	C, P	✗	H	H	• Additional overhead

Table 5: Access control summary with NDN compatibility. (C: consumer, P: producer, L: low, M: medium, H: high)

To this extent, some of the points to ponder are:

- In ICN, naming exists as a ‘shared’ layer in the middle of application and network layer.
- ICN has default structure for security which is centered at network layer. Such security structure can not only be used for securing content retrieval but also for securely retrieving certificates, access control rules and security keys.

4.1 Encryption Based Access Control:

In this access control scheme, generally the content creator encrypts the information and then defines certain rules for access the content. A requester must be directly authenticated by the producer. Some of the encryption-based access control mechanism as well as their sub-categories are discussed in detail as below:

4.1.1 Attribute-based Access Control:

Attribute-based Access Control abbreviated as (ABE) (Bethencourt and Waters, 2008), (Goyal and Waters, 2006), comprised of public-key encryption technique and the resulted secret key relies directly on the attributes set. The ABE has further two variants (a) Key-Policy Attribute-Based Encryption (KP-ABE) (Li and Shen, 2018) in which client has a private key which corresponds with access policy, on the hard hand the ciphertext is linked to some attributes (Lewko and Waters, 2011), (Lewko and Waters, 2008) (b) Cipher-text-policy Attribute-Based Encryption (CP-ABE) ciphertext is correlated to an access policy and the private key of client is associated with access policy (Lewko and Waters, 2010), (wang and Zhang, 2010). The major benefit of ABE is that it is flexible enough and can used to enforce access policies to a certain number of users (Belguith and Attia, 2018). ABE has been introduced with various variation and enhancement as AC mechanism for ICN. Some of them are discussed below:

Attributes-based Encryption and Routing (ACM SIGCOMM, 2013), an approach is proposed for content-centric privacy with reference of ICN model. The ABE is particularly associated with the data attributes and the access control policy is bind to the content.

Encryption Based control for ICN connected vehicles (T. Mahmoodi, 2017) an access control mechanism is designed which is named as EnCIRCLE for vehicular systems. Two main different modules are presented (a) Cryptographic module for content encryption subjected to access polices (b) an access management module for handling various polices and rules specified on the basis of their attributes.

A CP-ABE access control scheme based on hash table and data segmentation in NDN (Wu and Yue, 2019), this scheme introduces a scenario in which the producer divides the data into two halves: (a) Public Content Section (PCS) that is originally content itself: (b) Content Key Section (CKS) which includes key and contain data consumers. The requestor can only perform decryption by combining both PCS and CKS. While doing that, he needs to initiate two different interest packets one for requesting PCS from intermediate node or the producer. Second packet for CKS from producer. The main limitation of this scheme is that it always requires producer to be available.

4.1.2 Named-based Access Control:

Named-based Access Control (NAC) (Zhang and Zhang, 2017) system uses encryption algorithms that uses only content names and provides confidentiality of data and key management system.

Flexible End-to-End security in CCN (Wang and Li, 2015), the authors proposed an access control mechanism based on session (SAC). The proposed model utilizes two different names *public* and *secure* for a single chunk of information. Once a user is identified, he can identify secure names.

The demonstration of the model is done using an Online Social Network (OSN). Thus, a session is being established for an individual in order to communicate with an OSN while having a unique key for the services.

Data-based access control in Named Data Networking (Hamdane and G. el Fatmi, 2013), a model in which the data is secured by encryption and lock password. The concept of namespaces is used to classify the access rights which are being described with in an Access Control List (ACL). A randomly generated key is encrypted using public key which provides revocation technique.

4.1.3 Identity-based Access Control:

Identity based access control enables the requester to satisfy his identity either before requesting or before consuming the data. The owner of the data conducts the authentication before providing requested content. The identification of consumer is done at publisher level by using public or deducible keys.

A credential and encryption-based access control solution for named data networking (Hamdane and G. el Fatmi, 2015), proposal was given for access control based on encrypted credentials. The solution utilizes Access Control Manager (ACM) which is responsible for providing certified credentials to the users and on the basis of these credentials access rights are provided accordingly.

Rendezvous-based access control for information-centric architectures (Fotiou and Alzahrani, 2007), the authors introduced an access control delegation model used for publish-subscribe paradigm. It involves the rendezvous points (nodes that are responsible for data lookup), that are helping to enforce AC policies. The access control policy is created by the content owner, which is then protected with an ID. The owner broadcasts the information of the content to the RVs. If the consumer is in need to have the content, it initiates a subscription message towards RV in order to initialize authentication. The RV first checks if the requester's identity exists in authorized IDs. If yes, then RV informs the content producer to deliver it to the requester.

Fine-grained access control for named data networks (F. Tseng and Y. Wu, 2019), a fine-grained approach for access control is introduced for NDN architecture. It actually contained four entities: Content Creator, Consumer, Online shop and Key Generation Center (KGC).

4.1.4 Re-encryption-based Access Control:

Re-encryption strategy could conduct the encryption process more than one time in order to perform authentication or update rules for access. (Son and H. Oh, 2014)

A Cache-Aware Mechanism to Enforce Confidentiality, Track-ability and Access Policy Evolution in Content-Centric Networks (Mangili and Paraboschi, 2014), developed and extension for ICN model based on encryption in which the content is encrypted by the producer and the caching nodes stores the encrypted content instead of plaintext. The content producer itself authenticates the requesters and provides keys needed for decryption.

Achieving secure and scalable data access control in information-centric networking (Zheng and Azgin, 2015), authors introduced an idea of dual-phase encryption which integrates one-time decryption key, proxy re- encryption and all-or-nothing transformation all together. The content creator performs encryption of the original data with a key derived from private key of the creator. When subscriber requests particular information, the edge router further performs the encryption

with a random key of the data which is already encrypted by its content producer. In order to have full access to the content, the requestor must use both the keys (from original producer and edge router) for decryption.

Approach	Level	Content	Cache	O/H	Comp.	Major Drawbacks
Attribute-based Access Control						
• Attribute-based encryption mechanism	C	Dynamic	✓	H	L	• No evaluation presented • Missing revocation schemes
• Access control management based on attribute encryption	N	Static	✓	H	H	• Third-party authentication • Component placement and communicate models are missing
• Ciphertext policy using attribute encryption.	P	Static	✓	H	L	• Requires publisher to be always online.
Name-based Access Control						
• Session-based mechanism	P	Dynamic	✗	H	H	• Replicated content in the network
• Data-based access control using encryption and lock password	P	Static	✓	M	H	• Hard to generate/update access rules
Identity-based Access Control						
• Access control using certified credentials	P	Static	✗	M	M	• Use of a centralized entity • Not efficient in large-scale networks
• Fine-grained access control.	C, P	Static	✗	H	H	• Trustability issues

Approach	Level	Content	Cache	O/H	Comp.	Major Drawbacks
• Use of rendezvous points to enforce AC in pub-sub	P	Static	✗	H	H	• Heavy and intensive computation • Requires publisher to be always online
Re-encryption-based Access Control						
Update of access rules at content stores	P, R	Static	✓	H	H	• No guarantees in terms of updating all cached versions • Necessitates the availability of original producer
• Dual-phase encryption mechanism • One-time decryption key, proxy re-encryption, all-or-nothing transformation	P, R	Static	✓	H	H	• Scalability is an open question • Necessitates the availability of original producer • Trustability of edge routers

Table 6: Encryption-based Access Control strategies (N: Network C: Consumer, P: Producer, R: Router | L: Low, M: Medium, H: High)

4.2 Encryption-Independent Access Control:

The various existed encryption-independent schemes for access control are discussed blow. Further classification has been done for the proposed solutions.

4.2.1 Broker-Based Access Control:

In broker-based AC, an external entity is involved behaving as a broker which is responsible to implement and verify access control rules. The security of this method directly relay on trusting this broker.

Access Control Enforcement Delegation for Information-Centric Networking Architectures (Kutscher ACM, 2012), presented an AC scheme for ICN model to safeguard the privacy and information of the subscribers. This scheme protects the information of users constructed on AC policies that are formulated by Access Control Providers (ACPs). This ACP could be considered as an LDAP which ensures the requesters must satisfy access control policies. Every producer send its content to ACP to get the URI assigned. Then transmits the content to RNs. Now the ACP is in-charge of performing verification of the policy and confirms if the requestor has permission to access requested information. If yes, then RN provides the requested data to the user.

Time-based content access control mechanism for ICN subscription systems (T-CAM) (Zhu *et al.*, 2020), proposed a time-based solution to ensure the integrity of the data. In this method, the content owner encrypts the content using their identities also indicating the time for access. The approach uses three different methods namely proxy re-encryption, identity-based encryption and broadcast

encryption to ensure security of content distribution and access management.

A secure, efficient, and accountable edge-based access control framework for Information Centric Networks (Xue *et al.*, 2019), presented a model named SEAF that conduct authentication at the edge node of the network that has capability of blocking unauthorized request initially.

4.2.2 Cache Enforcement-based Access Control:

Caching mechanism is the core component of the ICN architecture that boosts up the entire network execution. However, this caching element works in a fashion such that the original owner is unaware of it, also the requester doesn't know from where the content is coming. This leads to the content owner losing control over the content causing copyright and privacy issues.

Copyright protection in named data networking (Tan and Chen, 2014), provided a solution for copyright issues while delivering the content to the requester in ICN architecture that arises due to caching feature. The authors claim that relying only on encryption will not solve content copyright issue. Therefore, they provided a solution by splitting the content into two portions big and small respectively. Such that without the smaller part, there is no chance of rebuilding the information. idea involves using of OR logic operation for rebuilding content.

Access-Controlled In-Network Processing of Named Data (Marxer and Tschudin, 2016), presented a prototype in which the data enforcement works in fully owner defined fashion. Every content piece contains an Access Control List (ACL) that holds the information of all the authorized users that can access the data. User sends the interest packet to have the desired content with the public key as an attachment. As soon as the publisher receive this interest packet, it checks the public key of the requester to find out an entry in the ACL list. The publisher then forwards the symmetrically encrypted data towards requester with the symmetric key which is encrypted asymmetrically with the public key of user.

A collaborative data access control scheme in named data networking: CDAC (Chongqing IEEE, 2019), presented a scheme in which access control for data is carried out at the caching routers instead of the original data owner end. The data is encrypted by the original owner with the help of symmetric key. After that the producer forwards the key with another alternate key particularly for the content. When a subscriber needs a desired content.

4.2.3 Interest-based Access Control:

Interest based Access Control formulation uses an additional information as an attachment with the interest message that helps in finding out the access rule contained for desired data. In this pattern, access control mechanism can be implemented on cached data.

Interest-Based Access Control for Content-Centric Networks (Ghali and Wood, 2015), presented an approach that uses an attached information with the interest packet to impose access control. The proposed scheme uses the two methods for security purposes hash- and encryption-based name puzzling. In this way, none of the unauthorized user can forecast the data. To counter the interest replay attacks mutual trust framework is being used between consumer and routers. This enables the caching nodes to conduct authorization before fulfilling the requests of desired content.

4.2.4 Signature-based Access Control:

Signature based access control method involves comprehensive procedure of signature confirmation for the sake of implementing access control polices.

Lightweight integrity verification and content access control for named data networking (LIVE) (Li and Fu, 2015), proposed an extension in NDN model that performs universal content signature verification. It uses one-way hash function which is consisted on Merkle Hash Tree algorithm that allows to produce signatures for content. Also to perform verification and sign various content pieces. This approach also permits data owners to manage access over the content by particularly distributing tokens only among the authorized users.

Approach	Level	Content	Cache	O/H	Comp.	Major Drawbacks
Broker-based Access Control						
• Use of AC Provider to coordinate access rules	N	Static	✗	H	H	• System security relies on the AC Provider • Requires extra overhead that may affect network delay
• Use of broker to store different access policies	N	Static	✗	H	H	• System security relies on the broker • Requires identification/verification of subscribers/publishers
• Edge-based access control framework	N	Static	✓	L	L	• Performance based on edge node
In-network Cache Enforcement Access Control						
• Enforces copyright when retrieving cached content	P, R	Static	✗	H	H	• Caching of split content is not feasible in real scenarios • Necessitates the availability of original producer
• Content-attendant policies in named function networks	C, P	Static	✗	H	H	• Extra communication overhead to identify consumers • Necessitates the availability of original producer
• Collaborative data access control scheme	C, P	Static	✓	H	H	• Overhead the network
Interest-based Access Control						
• Use of interest packets to enforce access rules	C, P	Static	✗	H	M	• Requires mutual trust between consumer and provider • Identical content with multiple names remains an open issue
Signature-based Access Control						
• Lightweight integrity verification	P, R	Static	✓	M	L	• Token refresh mechanism is required

Table 7: Encryption independent access control strategies (N: Network C: Consumer, P: Producer, R: Router | L: Low, M: Medium, H: High

Chapter 5

Conclusion

The next generation of internet demands huge dissemination of information. This new demand motivates researchers to find a solution that could fulfil this need. Information Centric Networking ICN came out as one of the solutions for this need that focuses on content itself and tends to satisfy coherent content delivery. There are multiple proposals for ICN framework such as NDN/CCN DONA, PURSUIT. ICN has certain characteristics that make it unique over currently deployed Host-centric networking. The prominent features of ICN are location independent naming, name-based routing and caching. This thesis work covers five main aspects about ICN. At first an overview of an ICN model including various ICN architectures was presented. A second, discussion about the existing Access Control mechanisms in ICN model was done. At third, detailed classification of ICN attacks has been presented along with their impacts into the system. Fourth, a connection is shown between ICN characteristics that may assist an attacker during an attack and after that severity of each attack has been depicted. Lastly, counter measures and a survey of proposed security solution for each class of attack has been discussed. Based on a survey of connection between ICN attacks, characteristics and security needs, the most affected parameters are availability and privacy in ICN model.

Availability: Sending huge number of malicious requests to disrupt the routing and caching systems. That are the core components of ICN architecture and in case of failure it will cause unavailability of the system.

Privacy: Accessing the interests will disclose all the information about requestor. Hence the user's privacy will be compromised.

ICN frameworks that employ name-based routing (CCN/NDN) to direct the requests within the network will perform better and attacks like DoS/DDoS will cause less damage. This is due to large network span of interests' request. In contrast the architectures that only route the requests to precise nodes to maintain efficiency (PURSUIT) will greatly impact the resilience of attack. Second lesson is that the security check must be done at the initial level. For instance, the attacks such as cache pollution and content poisoning are occurred due to insecure naming and caching mechanism. These attacks can be overcome by deploying solid caching and verification systems. As the data travels in plain-text during content retrieval within the network, this creates major threats for the data. There is solution provided that could furnish data security. But there is still need to do some major analysis to create a mechanism that can maintain privacy meanwhile leveraging all the benefits that ICN model offers. In this regard it is still not which framework would suit the privacy thus there is need to explore it in dept. While designing the security solution, the core concepts of ICN should be closely followed. ICN model encourages content availability by adopting ubiquitous caching mechanism thus it demands more advance resource independent access control systems. In this thesis work some of the existing access control mechanisms have been presented. However there still a need for comprehensive study to find out which access control system performs the best in which ICN architecture.

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